On Nicomachus' Introduction to Arithmetic

ΙΑΜΒΛΙΧΟΥ ΕΙΣ ΤΗΝ ΝΙΚΟΜΑΧΟΥ ΑΡΙΘΜΗΤΙΚΗΝ ΕΙΣΑΓΩΓΗΝ

Iamblichus

translated by Claude AI

Plato Scroll · ***plato-scroll.com***

Scan to read online, free and offline
plato-scroll.com/s/iamb-nicomachus/

On Nicomachus' Introduction to Arithmetic

Translated from the Greek text of Ermenegildo Pistelli (Leipzig, 1894). The numbers in the margin are the section numbers of the digital (First1KGreek) edition of Pistelli's text; they do not correspond to Pistelli's pages, the standard reference for this work.

1

In beginning our particular account of the mathematical sciences, distinguished as they are part by part, we begin from arithmetic; for the study that belongs to it is by nature the elder, because it deals with simpler and more primary things, and for this reason the account of it also precedes the other mathematical sciences. Now this account too is not simple but of many forms; for as many as are the genera of beings, it is divided along with them all and admits the same divisions.

2

But before the numbers that are contemplated in other things one must contemplate number itself by itself, and starting from this we shall be able to examine also the number that is in nature, or in moral characters, or in the forms, or in general in all the things that are. For this reason, then, we must take up the mathematical science of numbers. For indeed this must be laid down first, as in the place of a foundation; for when it has been laid down beforehand it is possible for the other sciences also to be acquired, but without it those too are not acquired. And with a view to learning, too, one must begin from here; for once the necessary theorems in the mathematical sciences have been determined beforehand, we are guided through them to the more perfect studies of numbers; for it is clear that those are in harmony with this one. As for the conception of it, it is right to set it before us not as consisting in bare notions, nor as arising subsequently upon sensible things, nor as stripping off and separating certain images from sensible things, but as able to fit common concepts to all numbers, in whatever way they subsist. It is about

mathematical arithmetic of this kind, then, that we now propose to speak.

3

Now we find that Nicomachus, in his Art of Arithmetic, has set forth everything concerning it in accordance with the mind of Pythagoras. For the man is great in the mathematical sciences and had as his guides in them those most experienced in mathematics; and, apart from this, he hands down with precision a wonderful order and theory and, together with wonderful demonstration, a knowledge of the scientific principles; he knows how to give an account of them, and he hands down the theorems pure and genuine, in no way muddied by alien opinions. Further, he is varied and copious, orderly and articulate in his knowledge of numbers, and universality of knowledge and inventiveness are present in him in great abundance; for he hunts out the first constitution and the first generation of numbers. He has also the quality of leaving nothing out; for he has gone over all the genera and species of numbers in common, has comprehended infinite things in finite ones and disordered things in ordered ones, proceeds through genera and species in order without passing over what comes next, and comprises in a single complete theorem what is circulated incompletely in many theorems.

4

Here he has also what one would least find present in him in his other books, conciseness and precision, and together with these fullness and completeness, a manner terse and compact, rich in thought and fertile —because, as I for my part believe, he sets down the Pythagorean mathematical doctrines on number themselves in their purity; but let anyone who wishes conjecture about this as he wishes. What must be concluded from all this, however, is the following. For if on all these grounds we prefer this man as the most skilled of all in arithmetic, it is with good reason that we for this cause set down his whole Art of Arithmetic, thinking that we ought neither to publish it incompletely, having lopped off its leading parts, nor to rewrite it—for this too

would be superfluous—nor to appropriate what has been written; for it is an act of the utmost ingratitude to rob the author of the reputation that falls to him. Nor yet, on this account, ought we to compose discourses foreign to the Pythagorean schools; for it is not our purpose to say new things, but the doctrines held by the men of old, and hence, neither taking anything away nor adding anything, we now set out in our discourse the Art of Nicomachus itself.

5

But in order that the present treatise may not be incomplete in this respect either: Pythagoras was the first to give philosophy its name, and he said that it is a desire for and, as it were, a love of wisdom, and that wisdom is knowledge of the truth in the things that are. By 'things that are' he understood and meant the immaterial, eternal and alone active things, that is, the incorporeals; while the rest are 'things that are' only homonymously, so called by participation in these—corporeal and material forms, subject to generation and destruction, and never truly being. Wisdom, he said, is knowledge of the things that are in the proper sense, and not of those that are so homonymously, since corporeal things are not even objects of knowledge and do not admit of secure cognition, being unlimited, not to be comprehended by science, as it were non-existent in contradistinction to the universals, and unable even to fall under a definition with any clear circumscription. But of things that are by nature not knowable it is not possible even to conceive a science; it is therefore not likely that there is desire for a science that does not subsist, but rather for the science concerned with the things that are in the proper sense, that always remain the same and in the same state, and that always coexist with their appellation. For indeed it happens that the apprehension of these is accompanied by that of the things that are homonymously, though the latter was never made an object of pursuit, just as knowledge of the particular accompanies knowledge of the universal; 'for accordingly,' says Archytas, 'having

discerned well concerning universals, they were bound to see well also concerning particulars, what they are like.'

6

Hence the things that are are neither of a single kind nor simple, but are at once seen to be varied and of many forms: both the intelligible and incorporeal things, whose appellation is 'things that are', and the corporeal things that have fallen under sense-perception, which by participation have a share in coming truly to be. It would follow that we should give a systematic account of absolutely all the things that are, in some such way as this. The whole nature of the continuous and of the discrete is conceived along with the things that are—that is, with the constitution of the entire cosmos—in two ways: that of the discrete by juxtaposition and heaping, that of the continuous by unification and mutual coherence. Properly speaking, the continuous and unified would be called magnitude, the juxtaposed and discrete, multitude. And in respect of the essence of magnitude, the cosmos would be conceived and spoken of as one, solid, spherical, grown together with itself, extended and cohering; while in respect of the form and notion of multitude, on the other hand, the arrangement and ordering and harmony of the universe would be conceived as having its constitution out of so many, let us say, elements and spheres and stars, genera and animals and plants, contrarieties and likenesses.

7

But of the unified, division is in every case to infinity, while increase is up to a definite limit; whereas of multitude, by a reciprocal relation, increase is to infinity, but division conversely is up to a definite limit—both thus being by nature, so far as conception goes, infinite, and for that reason not to be circumscribed by sciences; for 'there will not be anything at all that is going to be known, if all things are unlimited', according to Philolaus. But since it is necessary that the nature of science be discerned in things that have been so precisely fashioned by divine providence, certain sciences, cutting off a portion of each and setting a limit to what they had comprehended, called what was

taken from multitude 'quantity', which is thereby knowable, and what was taken from magnitude, in the same way, 'size'; and they brought both these genera under sciences, their own forms of knowledge—quantity under arithmetic, and size under geometry.

8

But since these were not of a single form, and each of them admitted a still more particular subdivision—for of quantity one kind was by itself, free from any relation to another, such as, for instance, even, odd, perfect, deficient and the like, and the other was in some way related to something else (which is specifically called relative quantity), such as equal, unequal, multiple, superparticular, superpartient and the like; and again of size one kind exists and is conceived as at rest, the other as in motion and revolution—for this reason, naturally, two other sciences joined the two already brought forward and took hold along with them of the study of each object of knowledge. For with arithmetic, which received as its own province the inquiry into quantity by itself, music came to share in the systematic treatment of relative quantity (for its harmonic part and its part concerning concords profess nothing other than to articulate the relations and ratios of the notes to one another and the quantity of their excesses and deficiencies); while to geometry, which is occupied with the examination of size that is at rest and stationary, spherics came as a helper, having been established as the judge of size in motion—of that size, namely, which is most perfect and has received an ordered and uniform motion.

9

Hence, since they are occupied with subjects that are sisters, it is reasonable to consider these sciences too as sisters, so that the saying of Archytas may not go unheeded, 'for these mathematical sciences seem to be sisters', and to regard them as holding on to one another in the manner of the links of a chain, and to hold that the kinship of these sciences ought to show itself as one, ending in a single bond, as the most divine Plato says, to him who learns in the proper manner;

and the man who has grasped all these things in this way, as he himself lays down, him he calls most truly the wisest, and he strongly affirms it in play; and he enjoins upon those who are eager to philosophize that these sciences are to be pursued and chosen by all means, whether they be difficult or easy—and very reasonably so, if indeed apprehensions of the continuous and the discrete come about through these alone, and the cosmos and all things in it are composed of the continuous and the discrete.[1]

10

Now the exact apprehension of quantity is wisdom, and philosophy is the striving after wisdom, and philosophy, alone of all the arts and sciences, secures for man the end that is proper to him and in accordance with nature, and leads to the happiness that belongs to him alone beyond the other animals and is pursued in accordance with nature, as the end most fit to be his aim. Of these four sciences, moreover, arithmetic plainly takes the lead, because it is found to be prior and more primordial. For it abolishes the rest along with itself, and again is implied along with them; and things that abolish others along with themselves but are not abolished along with them, or, put otherwise, that are implied along with others but do not themselves imply them, are shown to be in a way prior and elder. Hence the inquiry into the systematic treatment of arithmetic, as the very first, would be most reasonable and fitting.

10

Quantity—that is, number—Thales defined as a system of units (in accordance with the Egyptian view, since it was in Egypt that he pursued his studies); but the arithmetical one stands on its own, and so neither the unit nor the one will fall under these definitions.[2]

11

Pythagoras defined it as the extension and actualization of the seminal principles in the monad, or, put differently, as that which subsisted before all things in the divine intellect, from which and out of which all things have been arranged together and abide, numbered

out in an indissoluble order. Others of his followers defined it as a progression from the unit by the unit's own magnitude. Eudoxus the Pythagorean said, 'number is a definite multitude', distinguishing species and genus, just as quantity was distinguished above. The acousmatics of the circle of Hippasus said that number is the first paradigm of the making of the cosmos, and again the discriminating instrument of the god who fashions the cosmos. Philolaus says that number is the sovereign and self-generated bond of the eternal continuance of the things in the cosmos.

11

The monad is the least of quantity, or the first and common part of quantity, or the principle of quantity; or, as Thymaridas has it, limiting quantity, since the beginning and the end of each thing are called a limit, and of some things the middle as well, as for instance of the circle and the sphere.

12

The more recent writers define it as that in virtue of which each of the things that are is called one; but this definition lacked the words 'even if it be collective'. The followers of Chrysippus speak confusedly when they say 'the monad is multitude one'; for it alone is set in contradistinction to multitude. Some of the Pythagoreans said 'the monad is the boundary between number and parts'; for from it, as from a seed and an eternal root, the ratios increase in either direction reciprocally: those of the things that are cut to infinity growing smaller with an ever greater name, and those of the things that are increased to infinity, conversely, growing greater. Some defined the monad as the form of forms, on the ground that it contains potentially all the principles that are in number. For it is polygonal in the plane, from the triangle on to infinity, and solid, showing itself in all the forms, and spherical and conical, and recurrent, and a side and a diagonal, and, what is most general, heteromecic, whenever it potentially affords the notion of a greater arisen from itself, and

proportional and relational according to the ten relations, and in various other ways, in as many as will be pointed out.[3]

13

It is called monad (μονάς) from its abiding (ἐπιμένειν) throughout in its own principle; and so with all the other things too that are in this way given their principle by it.

13

Again, from another beginning: of quantity, by the first division, one part is even and the other odd. The even is that which is able to furnish from itself equal parts, both the greatest and the least: the greatest in size and in their relation to the whole, because it is divided into halves, and the least in quantity, because into two (for there are by nature no parts fewer than two, seeing that there is no number short of the dyad either; for this is the first system of units, which is the very definition of number in general); the odd is that which in every case, when it is divided into the fewest parts, yields parts unequal to one another. For it is not divisible in two into equal parts; for this would do away with the monad, which is by nature indivisible and which is of service for the whole systematic treatment and for inquiries into nature of this kind. And since the even number, in whatever way it is divided, whether into equal or into unequal parts, is in every case resolved into parts of the same kind, for both are either even or odd, whereas the odd is resolved into parts that differ, namely both the lengths of number, those of the school used to call the even 'heteromecic' from what thus befalls it, in accordance with the meaning of the word, as having in its partitions only the one of the lengths of number; and in contradistinction to this they called the odd 'amphimecic', as furnishing both of these together.

14

They may also be made known through one another in the natural series of number: the even is that which differs by a unit in either direction from an odd, and the odd that which differs by the same from an even. Each genus is given its form both properly and

accidentally: the even properly by the dyad, and accidentally by the unit as well; for the unit runs through it always in dyadic fashion, whether unmixed or with a difference as well, whether pure or together with any number whatever of the same kind; the odd, on the contrary, is properly measured by the unit, when it is measured an odd number of times, and accidentally by the dyad, not however by the dyad in itself, but together with the unit. The monad, however, has as its special prerogative over all the odd numbers, as being that which gives them their form, that it is not divided even into unequal parts; and the dyad over the even numbers, that it is divided only into equal parts. Hence the Pythagoreans used to name the monad Atropos and Apollo and other such names, and the dyad, analogously, Isis and Artemis. And from the fact that the monad is by nature indivisible, the same monad will be seen to be a limit and a demarcation in either direction.[4]

15

It is a limit for size, in order that from it, as from a whole, the division to infinity may begin; and for quantity, in order that in the same way the increase to infinity may be extended over against it, from it as a unit; and that from it as a whole there may arise a half, then a third, then a fourth, then a fifth, and so on, parts ever smaller and smaller, contrariwise to the increase of their names as it advances; while from it as a unit there arise a dyad, then a triad, then a tetrad, and so on in an advance without end, the increase following the names; and that from this there may spring up the generation of a manifold reciprocal denomination, the monad underlying both, size and quantity alike, in the manner of a joint, and making itself as it were a partition and boundary of their reciprocal denomination.[5]

16

For if we take the monad, and draw from it, as from an angle, a figure in the shape of a lambda, and fill up one of its sides in order with the numbers continuous with the unit, as far as we please, for example 2, 3, 4, 5, 6, 7, and so on to any extent, and the other, beginning from the

greatest of the parts, which is the half, the part nearest to the whole in magnitude, with parts that are likewise continuous in order, 1/3, 1/4, 1/5, 1/6, 1/7, and so on to any extent, we shall see the reciprocity of which we spoke, and a natural connection and well-ordered relation, such as the following.

17

Since the whole was divided into two, the part was named 'half' accordingly, and in this way the half was yoked with two; again, because it was divided into three, 'third', and into four, 'fourth', and so on in order as far as a hundredth and a thousandth and a ten-thousandth; and from this the necessity of division to infinity forces its way in, because it is extended alongside that which is admittedly capable of increase to infinity. And further, as twice one is two, so half-times one is a half; and as twice two is four, so half-times a half is a fourth; and as twice two twice, so half-times a half half-times: eight, and an eighth; and as twice three is six, so half-times a third is a sixth. And in general, whatever we take from either side, the ratio remains the same in it; and whatever holds without qualification in the case of each of the numbers will in every case be found, in proportion, in the case of the converse parts as well. But this must be grasped beforehand, as it will be useful for what follows: that, while all the parts answer in denomination to all the numbers, the half alone answers to two in fact, but not in name as well; for the language is wanting in this, as in many other things.

18

The generation of the odd is both from the unit and by an unbroken composition, not the cumulative kind but that by pairing, which some call 'conjugate': for example, first one, then 1 and 2, then again 2 and 3, and again 3 and 4, and so on in the same way. That of the even is by interweaving, as 1 and 3, 2 and 4, 3 and 5, 4 and 6, and so on to any extent, in order that the dyad, as that which gives the even its form and as its element, but not as actually even, may be left out; or, in another way, by each of the numbers from the unit being doubled, as

twice one and twice two and, in order, twice three, twice four, by which the giving of form to the even by the dyad, stated above, is made clearer. They may also be generated from one another, in the following way, so as to exhibit the property of number: for each of the two is half of the numbers of the other kind on either side of it taken together. And the most wonderful thing, which is both peculiar to the monad and goes to establish that it is not yet a number, is that, being bounded on one side only and not on both, it is the half of the dyad alone, contenting itself with its one neighbour.

19

Thus all the forms both of even and of odd are seen potentially in it in common, as in a kind of source and undifferentiated root of both, and one that is of necessity indivisible, unlike all the others. For against those who force the monad to be divided, and who set beside it on the other side the half, as a single quantity, continuous and of the same kind, there stand in the way the fact that, while all the numbers above it are yoked by their derived names with the parts severally opposed to them, it is itself set over against the whole alone; and the fact that there will in any case be a confusion of the two generic forms of number if we should say that the odd too is cut; and again the fact that it is possible rather to set beside it, on the side of the less, the nothing instead of the half, which in many places, even against our will, plainly admits itself into the nature of the theory: here, in the fact that the monad too is half of those on either side of it taken together, the dyad and the nothing, just as each of the remaining numbers was seen to be half of those on either side of it taken together; and there, far more and more plainly, when, 9 being the very first odd square after that which is so potentially, there appears in its mean, that is, in five, the principle of justice, making requital pair by pair according to arithmetic proportion, and just as the Pythagoreans define it when they say that justice is a power of rendering what is equal and fitting, contained in the mean of an odd square number.

20

For when the numbers from the unit up to the ennead are set out in a row, five, being the middle one, will mark off those on the near side of itself as having less than is fitting, and those beyond it as having more than their share, and that progressively: for those that come nearer to the ennead have always more, and those nearer to the unit always less; and what is fitting for each, by the principle of equality at any rate, is the ninth part of forty-five, the sum of them all, which is at once manifest in the mean between the more and the less, and in it alone, since justice too and the other virtues are found to be means between these and nothing else. For this reason, by as much as 9 exceeds what is due and has more than its share, by so much the first falls short; and by as much as 8, by so much the second; and by as much as 7, by so much 3; and by as much as 6, by so much 4; for with the gradual approach to the middle, as toward the suspension-cord of the beam of a balance, an equalization comes about, just as there too there comes about a rightness of the angles, both of those which the scale-pans make with the beam and of those which the beam makes with the suspension-cord itself.

21

But the middle number, 5, falls short by as much as it exceeds: by nothing, therefore. And this is one indication that the nothing is useful in the theory, and another shows itself at once. For not only is it in accord with this that 5 (ε) is, even in the shape of its written character, the half of 9 (θ), but also, further, that on account of their kinship the products paired on either side of it have by nature the same ending: for nine times 9 ends as once 1 does, eight times eight as twice two, seven times seven as three times 3, six times six as four times 4, and five times five alone ends as itself.

22

Further, nine times 5 ends as once 5 does, nine times 6 as once 4, nine times 7 as once 3, and nine times eight as once two. And again eight times 7 ends as twice 3, and eight times 6 as twice 4, and eight times 5 as twice 5, and seven times 6 as three times 4, and seven times 5 as

three times 5. And, in another way, six times 5 ends as four times 5, if not in name yet at least in force, just as we also showed that the half answers reciprocally in denomination to two in force, but not in name.

23

If, then, like just judges who render what is equal and due, we were to take from those that have more than their share and give back to those that have less, we shall not take at random from any chance number and give to any chance number, but according to the same proportion, using as gnomon and, as it were, as rule the one that has neither overreached nor been overreached, that is, the pentad; for this number alone justly possesses its own in full. From 9, then, we shall take the fifth number counted from it and give it to 1, and the one that has done the greatest wrong and the one that has suffered the greatest wrong will be made equal; and the fifth from 9 is 4, for there are 8, 7, 6, 5, 4. Again, from 8 we shall take away 3 and add it to 2; for the fifth from 8 is 3. And from 7 we shall take away the fifth from it, 2, and add it to 3, and they will be made equal.

24

And again, taking away from 6 the fifth from it, 1, we shall add it to 4, and they will be equal. But from 5 we shall take away nothing (for the fifth from it is nothing) and add it to 5 itself, and it will be equal to itself. Thus what is conceived as less than the monad, the monad being indivisible, namely nothing, everywhere preserves the proportion in relation to the monad, rather than the half, as those people supposed; and the monad too has itself turned out to be the half of the terms on either side of it added together, for the half of 2 and nothing is 1. Indeed the very name of nothing most expressively signifies to us that the monad is by nature least and indivisible; for "nothing" (οὐδέν), when divided as "not even one" (οὐδ' ἕν), takes away all being, which would not be so conceived if there existed a half of the monad, or a third, or the like parts of it. For why need one weave in besides the fact that the monad, when it multiplies any number whatever, does not go beyond that very number, seeing that

even when it does this to itself it does not depart from itself, as being by nature a boundary between number in the unqualified sense and nothing? For a number, whether it takes itself or another, brings the reckoning to rest in neither, but in every case begets some third number; whereas nothing, whether it should seem to multiply itself or something else, will never go beyond itself: for no times nothing, and no times 9, is nothing, since this is equivalent to "in no way 9"; and likewise in the other cases.

25

But the monad, as intermediate between the two, if it takes another number, leaves the reckoning in that number, and if it takes itself, in itself. And it must further be added, along with what has been shown besides, that advance answers reciprocally to descent and descent to advance. Thus once 9 is 9, and the reckoning has remained in the outermost extremes. And twice 9 is 18, and the reckoning has passed over to the second extremes; and so on in succession. For it belongs to another occasion to investigate further how, when the number is also arranged as a square from its exposition in a row, no fewer plausible results follow in addition, by nature and not by convention, as Philolaus says somewhere: the 5 being found here too, in like manner, to be the mean along the three rows taken now one way and now another, and only those that attend upon it in length and in breadth and also diagonally having received what is due to them, while those not so placed have more than their share or less, and that not at random, but according to a certain proportional reciprocity.

26

But for the present, referring the full account of these matters to the discussion specially devoted to justice, let us proceed to what comes next.

26

Of the even, then, by subdivision, one kind is even-times even, and another, its counterpart, even-times odd, these being as it were extremes; and intermediate between them and, so to speak, common

to both is the odd-times even. Not knowing this, Euclid and his school confusedly suppose the same number to be both odd-times even and even-times odd, having observed nothing with precision in this topic, most elegant though it is, as will be shown in what follows.

27

An even-times even number, then, is one that has its halves, and the halves of its halves, and further those of the parts below these, always even, as far as the unit; and for this reason it is an attribute of it alone to be measured by an even number only an even number of times. But if any number is, in addition to this, also measured an odd number of times by an even number, it will escape the description and will belong to one or the other of the remaining species. So here too Euclid again defines erroneously when he says: 'An even-times even number is one that is measured by an even number an even number of times'; for see, 24 is measured four times by 6, which is even, and six times by 4, and various others likewise, and yet they are not even-times even, not even on his own view; and a consequence of the even-times even is that both it and its parts and the parts of its parts admit resolution into two, and this as far as the unit, which is by nature indivisible. It seems to have obtained its name not only from being measured by an even number an even number of times, but also because every part that may be taken in it has an even name. And again the value contained in each part, that is, its units, is itself likewise even, in the same kind.

28

Its generation is the double ratio advancing in proportion from the unit to infinity. But if the even-times even numbers are set out from the root in an odd exposition, converging upon a single mean, the extremes among them will correspond reciprocally in name, and those next after them, and so on continuously as far as the terms beside the mean, so that the product of each pair is rendered equal to the square of the mean, since it alone, one and the same, answered to itself in name. But if they are set out in an even exposition, the

relation will be divided between two means that correspond reciprocally in name to each other, so that their product too is rendered equal to the product of the terms on either side of them, in regular order continually as far as the extremes. And in the advance of their generation the greater will in every case have, as their difference from the lesser, the lesser themselves, so that as a result the differences too, and again the differences of the differences, and those of these as far as is possible, keeping its ratio, are arranged in the manner of a triangle. And by their cumulative summation there arises in every case a generation of odd numbers, which will be useful to us a little later for the generation of the perfect numbers.

29

For in it the term that is about to come is always shown in advance, short by a unit, and all the terms to come are generically even, since the exposition consists of such numbers. And every even number, short by a unit, is necessarily odd. And in all expositions in proportion it is confirmed that the unit remains by nature indivisible; for it alone yields the appellation of the whole, which on each occasion corresponds reciprocally in name to the greatest term.

29

The even-times odd is the number which is itself also divided into two equal parts, in accordance with what is common to the even, yet does not have its parts further divisible, but each of them immediately odd; whence it also got its name, because, being even, it has its greatest parts immediately odd, or rather because the values of the parts in it are at variance with their names, being even when those have odd names, and odd when they have even names. And it was said to be opposed to the first species of the even not in this respect only, but also because of this species the greater extreme alone is divisible, and only once, being indeterminate and different at different times, whereas of that the lesser alone is indivisible, being determinate and always the same.

30

It is generated when the dyad multiplies the odd numbers in their order, so that, since the gnomons differ from one another by a dyad and that which multiplies them is also a dyad, the difference between the successive products is a tetrad; for this is twice two. And if we begin from the potentially odd, the result is the potentially even-times odd, 2; but if from the actually odd, 3, the actually even-times odd, 6. In the natural exposition of number, then, such numbers will be given their form by the dyad, will pass over three, will differ by a tetrad, and will be fifth from one another. And since the continuous, which is size, was seen to be the reciprocal counterpart of the discrete, that is, of quantity, and the former species has already made use of the proportion proper to size, this species too will make use of that proper to quantity, as being the opposite of the former; and so in accordance with the arithmetic mean the extremes added together will be equal to the means in an even exposition, and in an odd one to the mean taken together with itself, that is, they will be double of it, just as in the even-times even the square answered geometrically to the rectangles, where it resulted that the extremes, and the terms next within them as far as the middle, when multiplied by one another become equal to the product of the mean multiplied by itself, or else to the two means, these too being multiplied together just like the extremes on either side of them, in an even exposition of course.

31

And a property of this species, contrary to that of the former, is that it is measured only either by an even number an odd number of times or, conversely, by an odd number an even number of times.

31

But since at this point there is a more conspicuous error in Euclid, namely his failure to distinguish the even-times odd from the odd-times even, and to recognize that one of them is opposed to the even-times even while the other is a mixture of both, let us speak still more clearly about the third, first setting out Euclid's own words concerning them. For he speaks thus: 'An even-times odd number is

one that is measured by an even number an odd number of times.' And the same number is also odd-times even, for it is also measured by an odd number an even number of times, as, for example, 6: for if we say "twice three" it is even-times odd, and if "thrice two", odd-times even; very naively.

32

But in the third of his arithmetical books too he confounds the three into one, being evidently a slave to the suggestion of the name; for he says: 'If an even number has its half odd, it is both even-times odd and odd-times even', evidently saying the same thing as before. Then he adds: 'If an even number neither has its half odd nor is one of those doubled from the unit, the same number is even-times even and even-times odd and odd-times even.' So much for Euclid; but let us rather describe the third species as that which is in common moulded out of both, receives its form from both, and possesses their attributes. It is, then, a blend of them in its definition as well: for it is measured by an even number an even number of times, and the same number is measured by an even number an odd number of times, whereas this belongs to neither of the former both at once, but only the one attribute to the one and the other to the other. Moreover, its being divided more than once it has from the even-times even, and its not descending as far as the unit from the even-times odd; and in that the names of its parts are at variance with their values it shares with the second, while in that they also at the same time agree in name it is not cut off from the former; and on the side of the greater extreme, because it is bisected more than once, it runs to join the species that is bisected as far as the unit itself, departing from the one that is halved only once, whereas, in having besides the lesser extreme other parts too that are not resolvable, it to that extent departs from the former and becomes close to its opposite.[6]

33

Its generation too is a mixture from both. For one must set out all the gnomons of the even-times odd in succession from the triad, and the

even-times even numbers themselves on their own account, these being gnomons as well, in order from the tetrad; and then, starting from whichever side one pleases, for it makes no difference, one must multiply the first exposition term by term from the beginning, as far as one wishes, by the first term of the other, then the same terms again by the second, and after that by the third, then again by the fourth, and so on to infinity. For if the gnomons of the other are multiplied by those of the number that is divisible once, there will arise first numbers differing from one another by an ogdoad, doubles of the even-times odd, odd-times even numbers in breadth, in regular order from terms in regular order. Then, from a fresh beginning, come the doubles of these same numbers, quadruples of the original ones, having a difference quadruple of theirs, but necessarily double in relation to those before them; and so on throughout in the same proportion, the length too thereby coming into view.[7]

34

If, conversely, the gnomons of the even-times odd are multiplied by those of the even-times even, the same results will follow, but the length and the breadth will change places with one another, as though by exchange. In order, however, that it may be still more plainly apparent that Euclid was ignorant of these things, one must observe also, in the more extended expositions both in length and in breadth, that the attributes belonging to both of those kinds are present together in this one alone, as being a mixture of them; for it will employ the geometric proportion as the even-times even does, making the product of the extremes equal to the square of the mean or to the product of the means, according to the number of terms set out, and the arithmetic proportion, rendering the sum of the terms that enclose the mean or the means equal either to twice the one or to the two taken once. Thus in all respects this number is shown to belong in common to both and to be, as it were, their offspring, the species of the even that came before it being opposed to one another, since it neither differs altogether from either nor is altogether the

same. We must therefore correct Euclid's definitions and say that the number which is measured by an even number only an odd number of times is even-times odd, whereas the one that never has only the one character but always, of necessity, both, which neither of those has, is odd-times even, the two being together in it, reasonably a blend of both, and standing apart from the one by participation in the other that remains.[8]

35

Of the odd number, again, by subdivision, one kind is prime and incomposite and the other secondary and composite; and, in another way, one kind is prime in itself, which is thereby at once prime and incomposite relative to another as well, and the other is secondary in itself, which need not also be secondary relative to another; rather, of this latter itself one kind is prime relative to another, while the other will be secondary and composite relative to another as well.

36

A prime and incomposite number, then, is an odd number which is measured completely by the unit alone and not by any other part besides; and such a number will advance in one dimension only. For this reason some also call it 'euthymetric', and Thymaridas calls it 'rectilinear' as well; for it is without breadth in its exposition, being extended in one direction only. It is peculiar to it to have no part except only the one that bears its own name, the magnitude of which is of necessity a unit. It is called prime not only because its measure is the unit alone and no other number (and the unit is the very first thing and the element of number), but also because no number can be observed before it, none at least that is a system of units, of which it will itself be a multiple; rather, it will plainly offer itself as the first, for certain other numbers to become multiples of it. And it is called incomposite because it could not be resolved into numbers equal to one another, from which it is clear that it was not compounded out of such numbers either.

37

Secondary and composite is the number that has the characteristics opposite to those of the one just described: a part besides the one that bears its own name, either one or more, and in the same way a measure besides the unit, either one or more. Such a number, in addition to being measured out linearly in a straight line, will also be set out as a plane, either as a square, if it has one part besides the one that bears its own name, or as a parallelogram, if it has at all events two parts answering reciprocally to one another, owing to the difference of its sides. More parts might be found in either case in a multiple, when the exposition has been made an odd number of times as far as the original numbers. It is called secondary because it employs also some second measure, or even several, besides the unit, and because among multiples it is never ranked first but, correspondingly, after a prime or primes; and composite because it can also be resolved into equal numbers, from which it is evident that it was also compounded out of such numbers.

38

Starting afresh: of the secondary species, one kind is secondary and composite both in itself and relative to another, as 9 relative to 15 or 21, and the other is secondary in itself but prime relative to another, as 9 relative to 25 or 35. For these numbers, taken by themselves, employ other measures apart from the unit, but relative to one another they employ it alone. We must reject those who say that, conversely, there is a number which is prime and incomposite in itself but secondary and composite relative to another; for they are deceived through comparing the measure itself with the thing measured, and they do not see that the numbers must have a common measure other than the unit and other than both of themselves. If this befalls any number relative to another, that number, being secondary in itself as well, will also be secondary relative to another. But it is possible, on the contrary, for a number that is secondary in itself not to be so relative to another. If any two odd numbers are proposed for the determination of whether they are prime or secondary to one

another and, if secondary, what their common measure is, we shall continually subtract in turn the lesser from the greater as many times as possible, and the remainder from the original lesser number, and so on always in the same way, until the process ends either in a unit or in some other number from which it is no longer possible to subtract; and this number would be the common measure of the original numbers, which will then be called secondary to one another, as 15 and 35; for their common measure is the pentad.

39

But the unit shows them to be prime to one another and incomposite, whenever the process ends in it; for of such numbers it alone is the common measure.

39

In order that we may know how to generate in order all the numbers that are secondary and composite both in themselves and relative to one another, together with their measures and the correspondingly named parts, however many there may be, one must know a method of the following kind, which, like a sieve, will retain such numbers within the reckoning and will separate off the rest, namely the prime and incomposite, as things cast out. Set out in a row, in good order and in succession, the odd numbers from the triad, to the greatest length you possibly can; and, attempting to measure with the first, you will be able to measure completely, among those that follow, the numbers that leave two between them, without limit; with the second, those that leave four between them; with the third, those that leave six, and with the fourth those that leave eight; and, in general, with each number, those that leave between them twice as many as its own place in the order.

40

From this it is evident that each will measure, in accordance with its own name, the numbers that stand at the distance named after it, as 3, passing over two, always measures the third numbers, and so on consistently. But the first will measure the number first measured

after it three times, in accordance with its own magnitude; the one after that five times, in accordance with the magnitude of the next number; the one following that in accordance with the magnitude of the third; and so on in like manner throughout; while the second, taking over this function in its turn, will measure the fifth number from itself by the magnitude of the number before it, the fifth again from that one by its own magnitude, the next fifth again by the magnitude of the number after it, and so on without end; and similarly in the case of the rest. And the notion of the potentially odd, that is, of the unit, will show itself here too, whenever each of the numbers set out, on taking over the measuring and multiplying itself, makes a square, as 9 from three times three. In such numbers there is sameness, which derives from the monad, as in once 9; but since otherness derives from the dyad, it is reasonable that the numbers produced from different numbers multiplying one another will also have their sides different, answering to one another according to the magnitudes of the gnomons, and such a number will be called oblong.

41

For the sake of clarity: how many times they measure will appear from the exposition of the odd numbers from the triad onward without limit, and how many numbers they leave between from that of the even numbers from the dyad (this too is a token of the everlastingness and mutual affection of the two species of number, even if they seem to be contraries, as right is to left, and yet in the same way helpful to one another), or indeed from the doubling of the place of each, in which the measuring number is ranked. The numbers, then, that are marked by these measurings are plainly secondary and composite, and their common measure is the number that has come upon them; while those that are passed over, like what is cast out through a sieve, are prime and incomposite.

42

Here too Euclid falls into a most manifest error in supposing the dyad to be one of the prime and incomposite numbers, because it employs

the unit alone as its measure, having forgotten that it belongs to the species of the even, though it is odd-like in order that it may hold potentially, in a seminal manner, the principles of the kindred even-times even and even-times odd numbers, just as the monad holds those of all numbers without qualification; whereas the prime and incomposite numbers were seen to belong, by subdivision, to the odd species alone, and not to the even as well. At any rate he could not produce another even number of this kind, not even if he strained himself, so far is the one species of number by nature free of such a thing, just as the remaining species is free of the subdivisions of the former, the even-times even, the even-times odd and the odd-times even. But the dyad itself also, being as it were elemental and seminal, does not clearly partake of these subdivisions, although for them it is the leader of this genus, just as, of course, in other cases too the principles do not partake of many things that necessarily belong to the compounds: as the attributes of a line are not observed in a point, nor those of an interval in a note, nor those of a proportion in a ratio, nor corporeal attributes in matter and form, and as the attributes of many other systems, both drugs and mixtures, go beyond those of their several elements.

43

Again, starting from the beginning: of the even number taken by itself and here too entirely freed from entanglement with the odd, one kind is superabundant and another deficient, these being contrary to one another, while common to them and, as it were, a mean is the so-called perfect, differing in some respect from both and again in some respect partaking of both. It is superabundant, then, when an even number yields all its own parts, when added together, greater than itself and exceeding it in quantity; for this is why it has been so named, as being something faulty in its members, over-membered and overreaching, ranked in the class, as it were, of doing wrong and of having something more than falls to its share, as if someone had too many fingers on one hand or on a foot. It is deficient when, likewise,

an even number, compared with all its own parts added together, is found to be greater, and makes its parts less than itself, which is why it has been so named, being deprived of the parts that properly go to complete it, like someone who is overreached, in suffering wrong and not having received what is his own, as if someone were tongueless or one-handed.

44

An example of the former is 12 and its multiples without limit, and 18 and 20 and many others of the kind; of the second, 8 and 10 and 14 and the like.

44

Perfect is that which is observed as intermediate between these and which has its own parts, when added together, neither more, like the superabundant, nor fewer, like the deficient, but midway between the greater and the less, which is to say equal, as it were by a kind of justice and receipt of what is its own and proper to it. Such examples accord with the correct view that the virtues are certain moderations and means between excess and deficiency, and not extremes, as some have supposed them to be; and that evil is opposed to evil, and both together to a single good, while good is no longer opposed to good, but to two evils at once: as cowardice is opposed to rashness, the common character of which is unmanliness, and both together to courage; and knavery to stupidity, the common character of which is folly, and both together to prudence; and prodigality to avarice, the common character of which is illiberality, and both together to liberality; and bashfulness to shamelessness, the common character of which is want of modesty, and both together to modesty; and in the case of the remaining virtues and worthy dispositions the analogous thing will become apparent to us if we attend to it, just as in the case of the relation of the unequal too it will be shown that greaterness is opposed to lessness, the common character of which is inequality, and both together to equality.

45

On this account, then, the rarity of the perfect number, as of something good and not of evil, which is abundant, will furnish us, by a law of nature, with only one among the units, that is, within the decad, and only one among the tens, that is, before the passage into * * * among the hundreds, and only one among the thousands.[9]

46

And likewise, as it may be, only one in the first rank of the myriads, and again one in the second, and so on to infinity. Examples of this are 6 and 28 and 496 and 8128, and those like them, which end alternately in a hexad and an ogdoad. The method of their generation, too, is one that establishes the mutual affection of the species of number and their everlastingness in concord. For one must heap up the numbers that proceed from the unit in double proportion, that is to say the even-times even, adding them always one at a time, and examine the sum at the addition of each number. If a prime and incomposite number should result from the summation, let us multiply the number that has resulted by the last term taken in the addition; for the number so produced will in every case be perfect. But if it is secondary and composite, let us pass it over and add on the next term in the proportion, to see whether the number that results is prime and incomposite; for if it is, it must be multiplied by the term that was last added, and in this way the perfect number that is next in order will come to light. And so on without end. Thus through the addition of the even-times even numbers the nature of the even is indicated, and through the generation of odd numbers out of them, and above all through the production of the prime and incomposite, that of the odd.

47

One ought not to be surprised if a variety of things is predicated of the same number, as, for instance, of this very number 6, that it is perfect, and the first even-times odd, and again the first heteromecic, and that it is moreover called marriage by the Pythagoreans, because in it first of all there comes about a union of male and female by blending; and indeed on the same ground they call the same number health, and

also beauty, on account of the completeness and symmetry of the parts in it. But those have misunderstood who suppose that they also call the same number friendship, on account of the coming together and befriending in it of things that differ; for it is certain other numbers that they expressly call friendly numbers, in their assigning of the virtues and the worthy dispositions to the numbers, such as 284 and 220; for the parts of each of them are generative of one another, in accordance with the account of friendship, as Pythagoras declared; for when someone asked 'What is a friend?' he said 'Another I', which is shown in the case of these numbers.

48

But since we shall set out in its proper place what the Pythagoreans refer to this sort of study, which is a most flowery and elegant one, we must proceed to what comes next.

48

For it follows upon these matters to treat of quantity that is no longer in itself but now relative to something, not because the whole technical treatment of quantity in itself has reached its end (for how could it, when we have discussed neither the plane numbers, which are of every variety, nor the solids?), but because these numbers contribute most of all to the understanding of those. For indeed we shall not complete the account of these in one continuous course either, but, aiming at the ease that the order affords the beginner, we shall for the most part deal with them after those, having deferred in turn the exposition of the proportions. What, then, a little earlier appeared to bear upon the account of the virtues, in the distinguishing of the perfect numbers and their contraries, this same thing shows itself to us again at the very beginning of relative quantity.

49

For of numbers considered as standing in some way in relation to another, the two most generic relations are equality and inequality; and equality, like a kind of moderation and mean, is undivided,

admitting neither slackening nor intensification, whereas inequality is at its first division split into two, into the greater and the less, just as in the case of the virtues too their opposite, vice, was distinguished into excess and deficiency. The greater is opposed to the less, and both together to the equal; and nothing could be equal without being equal to something, nor greater or less without being so than something, which is why they are with good reason relatives. But to the equal the same name answers in return, as befits a mean, which is shown also in certain other examples of the relative, in the case of brother and fellow-soldier and neighbour and age-mate and others like them; whereas with the unequal something similar holds as regards the genus, but no longer as regards its species: there the answering term comes under a different name, as in other cases, such as father and teacher and beloved and the like. Equal, then, is a quantity which, when examined against its counterpart, has nothing more and nothing less; unequal is one which, when likewise examined against its counterpart, is either greater or less; for in this pairing the measure will leave something over in the thing measured after the single measuring.[10]

50

And the greater is that which, when measured by the other, is of such a nature as to leave after a single application some part of itself, of whatever amount, unmeasured; the less is that which, being the measure of its counterpart, cannot encompass the whole of it in a single application. By subdivision these two species of the unequal yield five relations each, and both together ten. For of the greater, one kind is the multiple, another the superparticular, another the superpartient, and the remaining two, when the multiple is mixed with each of the others, are the multiple superparticular and the multiple superpartient; and of the less, by reciprocity and with the prefix 'sub', one kind is the submultiple, another the subsuperparticular, another the subsuperpartient, and the remaining two, mixed as in the former species from the multiple and each of the

others, are the submultiple superparticular and the submultiple superpartient. The prefix in the names equips us to recognise the former as antecedents, as being first in nature and in dignity, as will be shown, and the second, which sink beneath them, as consequents, possessing the contrary characteristics.

51

If anyone should say that equality is not a relation, on the ground that the terms in it are without interval and without difference, he must be reminded that a relation is something other than an interval. For observe: in any chance example of inequality the interval between two terms is the same even if they are reversed, and yet when they are reversed the ratio, that is, the relation, is in every case different; so that nothing prevents the terms in equality from having no difference, being without interval, and yet in every case having a relation; otherwise the equal will not be among the relatives, which is impossible.

51

The multiple, then, is the first species of the greater, when of two terms the one measures the other more than once, and completely. It will begin from twice, so that from this they may take their names, the measured term being called double and the measuring term subdouble and, synonymously, half, just as, indeed, the remaining genus itself is called both submultiple and, synonymously, simply part; and if it is three times, the greater is triple and the less subtriple and third, and so with the other species in succession.

52

We shall have a clear illustration of all the multiples in their regular order if, having set out the continuous series of number from the unit, we compare either with the unit itself those that come after it, each one in succession, or with the dyad that follows it those that come after that, skipping one, each likewise in succession, with the triad those that skip two, or with the tetrad those that skip three, and so on to infinity, the multiples advancing in step with the setting out of

number to whatever length it is carried. And if we write down in parallel rows all the species of the multiple, beginning from the unit, setting out in addition the successive series of number and generating the multiple relation in depth in relation to it, we shall observe many other delightful consequences and a varied elegance, and in particular an orderly generation of the corresponding naming of superparticulars of every kind after multiples of every kind, according to their kinship, and of superpartients as well and, if one should look into it, of the mixed kinds too; and whole rows will be seen to stand to whole rows in one unvarying relation that advances in regular order, correspondingly both in breadth and in depth.

53

Moreover the roots of the superparticulars will be found in a single row in depth, the second from the root in the next, and the third and fourth according to their corresponding sequence with these, having as their differences the numbers in succession from the unit. And if we also remove the units along the breadth, as having nothing varied about them, and set in front in their place the continuous series of number under the same unit, we shall observe a certain elegance and, showing itself in seminal form, the principle of the method of the divinatory tablets, which is handed down among the epanthemata of the arithmetical introduction. And if the setting out of the multiples should be carried as far as the decad, there will be units at the corners both in length and in breadth, the extremes once and the middle one twice, so that here too the property of proportion is preserved; for the product of the extremes will be equal to the square on the mean, and one of the extreme units will have the role of a point, the other that of a square, and the middle one that of a side.

54

And whichever of the numbers in the diagram be taken, it will be half of the two on either side of it, both in length and in breadth. But if they are taken diagonally, in the one direction the middle term will be less than the half by a unit, and in the other greater by a unit.

Moreover, from the corner at the beginning, that is, the unit, to the corner at the end, the diagonal will consist of squares alone, each flanked by two heteromecic numbers both in length and in breadth, so that here too that universal rule is preserved, that from two successive heteromecic numbers and twice the square that is the mean proportional between them a square is in every case generated, and conversely from two squares and twice the heteromecic number that is the mean proportional between them likewise a square, and in the one case odd squares, in the other even. Now the production of even squares occurs in the present case too, because it is the squares alone that are flanked, which are by nature alternately odd and even, and the heteromecic numbers that attend them are always even, and the middle term, whether it be even or odd, makes an even number when it is taken twice; but the production of odd squares no longer occurs, since the heteromecic numbers are not flanked by squares; for when the extremes are taken once, among which there is in every case an odd number, the oddness persists.

55

And at each square, on either side in the shape of a gamma, the relations are again observed in regular order from the beginning, that is, from the double. And if we should also flank the heteromecic numbers in the shape of a gamma with the squares, adding the extremes once and the middle term twice, we shall produce the odd squares which we said were here left out.

56

The diagonal numbers will have as their difference from one another, in the one direction, the odd numbers from the triad, from beginning to end, and in the other, the even numbers from the dyad, from the middle to the extremities, the well-ordered numbers on either side being paired with one another in equality.

56

A superparticular ratio arises, generically, when of the terms compared the greater contains the other and in addition one part of

it; specifically, if the part is a half the ratio is sesquialter, if a third sesquitertian, if a fourth sesquiquartan, and so on in sequence always, the greater terms becoming antecedents in relation to the lesser, and conversely the lesser consequents in relation to the greater, these too always taking their name with the prefix 'sub'. An example of them is had, for the sesquialter, if, when the continuous series of number has been set out, we pick out the even numbers from the dyad and compare with the first the number with none intervening, with the next the number with one intervening, with the third the number with two intervening, and with the fourth the number with three intervening, and so on in sequence.

57

For the sesquitertian it is had when we pick out the numbers from the triad that differ by a triad and compare with them, with the first the number with none intervening, with the second the number with one intervening, with the third the number with two intervening, with the fourth the number with three intervening, and so on in sequence as in the former case. We shall have an example of the sesquiquartan if we pick out the numbers from the tetrad that differ by a tetrad and again compare with them, with the first the number with none intervening, with the second the number with one intervening, and with the third the number with two intervening, and so always in the same way as in the cases already described. In the remaining species of the superparticular too we shall do what is analogous, taking, in accordance with the very name of the part, the first numbers that are able to furnish from themselves the part by virtue of which the numbers compared with them will be their superparticulars; and these latter will also differ from them by a unit and will become the roots of the ratios. The designation of the part, which is always considered with reference to the lesser term, will be greater in name by a unit with reference to the greater. But the examination of the part will not be made with reference to the greater terms, because none of the root terms will be found to possess that part by virtue of

which each of them is the superparticular of the lesser term compared with it; and it is according to the roots that the ratios increase.[11]

58

A superpartient relation exists when the greater term contains the lesser and in addition certain parts of it, more than one, obviously. If these are two, it is called superbipartient and the lesser term subsuperbipartient; if three, supertripartient and subsupertripartient; if four, superquadripartient and subsuperquadripartient, and so on in sequence. We shall have an example of superbipartients if, setting out the odd numbers from the triad, we compare with each of them the number with one intervening, and of supertripartients if, setting out the continuous numbers from the tetrad, we compare with them the numbers with two intervening. But since formations of this kind are not pure but are confused with other relations, we shall use the progressions according to the ratio of multiples, taking as roots, just as in the case of the single parts, the terms that will furnish from themselves the parts after which the superpartient is named: for example, of superbipartients, five to three, then the doubles and triples of these and so on to infinity; of supertripartients, seven to four, then the doubles and triples of these and so on in sequence; of superquadripartients, nine to five, and analogously throughout, so that the progression of the lesser terms in the roots goes by the successive numbers from the triad, and that of the greater terms by the odd numbers from the pentad.

59

In general we shall have the roots of every ratio as follows. In multiples, they are those in which the unit is the lesser term of the terms compared, and it is a special feature of the double that the same unit is also the difference. In superparticulars, in the sesquialter the dyad will be the lesser term, and the terms will again have a unit as their difference. In the sesquitertian and the sesquiquartan and the succeeding superparticular ratios the lesser term will be the one that furnishes from itself the name for the part by virtue of which the ratio

is superparticular, and the difference in all of them will be the same unit.

60

But in the roots of the superpartient too the same unit, although it has no place either to show itself in the terms, as in the species of the multiple, or to be their difference, as in those of the superparticular, because the greater term exceeds the lesser by more parts than one, may be seen in the terms in another manner; for the parts left over unmeasured in the greater term, when compared with the lesser, will in every case have a unit as their difference.

60

It remains to speak of the mixed relations compounded of the multiple and the remaining two, the superparticular and the superpartient, and of the consequents of these, so that the relations of inequality too may have their generation naturally in accordance with the perfection of the decad, the antecedents being five, and five the consequents conjugate to them: the antecedents in the multiple, the superparticular, the superpartient, the multiple superparticular and the multiple superpartient, and the consequents, equal in number, named with the prefix 'sub'. For the relation of equality, inasmuch as it has no difference but is as it were sameness and unity, since the equal is as one to one, will be of another nature, and indeed of the nature contrary to inequality, and for this reason will not be numbered together with the species of inequality.[12]

61

Moreover, equality will have the role of a principle in relation to inequality, just as among linear things the right angle stands to the obtuse and the acute, and among musical intervals the mese to the notes that are tightened and slackened. For these too, taking their start from something definite and limited in accordance with the principle of equality, and from this having their deviation toward the greater and the less in accordance with inequality, proceed to infinity. In order, then, that it may be shown that the relations of inequality

arise from equality by natural necessity and not by our own positing, first the multiple relation, beginning from the double, from which in turn comes the superparticular, taking its beginning from the sesquialter, and from this the superpartient in the analogous order, and next from these the mixed relations, one must set out three terms, first in units, then in dyads, and again in triads, and so on in sequence, and alongside each such setting-out one must fashion three other terms by means of three rules that are always the same, and alongside each set of terms so fashioned three more, and from these others again, and so on always in sequence.[13]

62

At each formation one must try setting out the terms both in their natural order and in reverse, and, using the same rules, fashion as a second setting-out the terms derived from them. The rules will be these: make the first term equal to the first of those set out, the second equal to the first and the second together, and the third equal to the first, two seconds and the third together. From all sets of three terms, then, set out beforehand in equality, whether in units or in dyads or again in triads and so on, there will arise by the aforesaid rules generically multiples, and specifically, among multiples, the doubles: the first from units, the next in order from dyads, those after them from triads, and so on in sequence. From the doubles so fashioned come triples, the first again from the first and the next from the next; from the triples, quadruples, preserving their own good order; from the quadruples, quintuples; and always the succeeding ratios from those that precede. But if in fashioning them we do not use this setting-out of the terms in their natural order, but reverse the first terms fashioned from the equalities, so as to place the third term in the position of the first and the first in that of the third, while keeping the mean term likewise as mean, and then fashion others by the same rules, there will spring up generically superparticulars from multiples, and specifically sesquialters from doubles, well-ordered from well-ordered, sesquitertians from triples, preserving the same

order, and likewise sesquiquartans from quadruples and sesquiquintans from quintuples and so on, the superparticulars that share the name of each species extending side by side with the species of the multiple by a certain natural kinship.

63

And from these same terms again, when they are reversed, we shall in every case generate the superpartient ratios, the first again from the first, the second from the second, the third from the third, and so on in sequence, these too being disposed in good order in correspondence with the original derivation of their names. For the sake of an example, let three units be set out beforehand in the ratio of equality. If, then, in accordance with the rules stated we make the first term equal to the first, it will be a unit; if the second equal to the first and the second, it will be a dyad; if the third equal to the first, two seconds and the third, it will be a tetrad; and the terms so fashioned will be in double ratio, 1, 2, 4.

64

From these, by the same rules, we shall have the terms in triple ratio, 1, 3, 9, and from these the next after them, in quadruple ratio, 1, 4, 16, and so on in sequence. If we set out dyads beforehand in equality, the next terms, being likewise in double ratio, will be 2, 4, 8; from these again the next, the triples 2, 6, 18; from which the next, the quadruples 2, 8, 32; and all the rest follow together. But if we reverse the first terms in double ratio, 1, 2, 4, we shall by the same rules produce the first terms that are in sesquialter proportion, 4, 6, 9, and from these in turn, when reversed, the terms that are likewise in superbipartient proportion, 9, 15, 25.

65

From this the kinship of the relations becomes plain. For if the double ratio was generated from equality, and we have learned that the half takes its name from two, it is with good reason that next, as akin to it, the sesquialter ratio was fashioned among the superparticulars, and from this in turn, among the superpartients, in accordance with its

affinity with the dyad, the superbipartient. If the first terms are in triple ratio, sesquitertians will grow out of them and from these supertripartients; if they are quadruples, sesquiquartans and superquadripartients, and so always with those that follow, preserving the affinity of the derived name, roots coming from roots, seconds from seconds, thirds from thirds, and always in the same way. But let us not suppose that the roots of superparticulars in three terms will be the same as those that appear in two; for when a given ratio exists in two terms it is not possible for a third term also to be procured that preserves the same ratio to the mean term, because the greater term is not capable of furnishing the same part by virtue of which it is the superparticular of the first, so that the third term too might by that part preserve the ratio to it.

66

For every root of a superparticular ratio, having its terms differing by a unit, will not have them divisible in like manner; but if the lesser is divided into two, the greater is divided into three, and if the lesser into three, the greater into four, and always the greater will admit a division greater in name by a unit than that of the lesser, so that, since in any ratio whatever the part is examined with reference to the lesser term, which is consequent in relation to the greater, there will be no third term as antecedent having the greater as its consequent according to that part. But since the roots in three terms are not the same as those in two, the roots will show themselves in the proportional terms in another way, for they will turn out to be their differences. For instance, since 4, 6, 9 are proportional in sesquialter ratio, their differences will be the roots that contain the same ratio, 3 and 2; and again, 9, 12, 16 being in sesquitertian ratio, the differences of these will be the root terms, 4 to 3; and always in the same way the same thing will happen in all the species of the superparticulars; for those will be roots in three terms whose differences are the roots in two.

67

In the multiples, the proportional terms set out from the beginning will always have their lesser terms as roots in each ratio. The cause of this is the unit, which offers itself as consequent to all the ratios of the multiple. None the less the differences in the proportional terms will also contain the same ratio, even though they are not roots of the ratios, as they turned out to be in the case of the superparticulars. Only the proportional terms in the double ratio from the beginning will have this as a special privilege, that they also have as their differences the lesser terms, which are precisely the root terms. In the species of the superpartients, the terms containing the roots of the ratios will appear neither in the differences, as with the superparticulars, nor in the lesser terms, as with the multiples, but according to a certain other well-ordered proportion. For the terms that are proportional in the superbipartient ratio will contain the root terms in the half of their differences, the kinship of the half with the dyad, in virtue of which the ratio is superbipartient, showing itself here once again; those in the supertripartient ratio will contain them in the third of the differences, those in the superquadripartient in the fourth, and those in the superquintipartient in the fifth, and the like will hold always in succession, the natural affinity of the fractional part with the ratio being preserved.

68

For the ratios taken by themselves too have, in respect of the parts, a name that is less by a unit when they are compared with the fractional parts in which the excess of the greater term over the lesser consists: for the first ratio will be superbipartient of thirds, the second supertripartient of fourths, the third superquadripartient of fifths, and so on in like manner.

68

The mixed relations, compounded of the multiple and of each of the remaining two, the superparticular and the superpartient, are themselves also generated from those that precede them: the one in multiple superparticular ratio from the one in superparticular ratio,

from which the one in superpartient ratio was also generated; for example, specifically, the double sesquialter springs from the relation in sesquialters, the terms no longer being set in reverse order but in their natural order, while we make use of the same three rules: for, there being a proportion in sesquialter ratio, 4, 6, 9, whose differences are the root terms, the double sesquialter will be formed in the terms 4, 10, 25. From the proportion in sesquitertian ratio, 9, 12, 16, whose differences again are the root terms, when we begin in the same way from the lesser term, there comes the double sesquitertian in the terms 9, 21, 49.

69

And from the proportion in sesquiquartan ratio, 16, 20, 25, whose differences are again the root terms, the double sesquiquartan is generated in the terms 16, 36, 81, and so on in like manner, the kinship being preserved here too between the fractional part that runs on after the multiple and the name of the superparticular ratio from which the mixed relation has its generation. For when the generating relation is sesquialter, the one generated is double sesquialter; when it is sesquitertian, double sesquitertian; and when it is sesquiquartan, double sesquiquartan; and so on in sequence. Again, the roots of these too will be found arranged in good order, no longer showing themselves immediately in the differences of the terms that are formed, as happened with the simple relations; rather, because the relations are mixed and the ratios have been increased, they will be seen to lie in fractional parts of the differences.

70

For the root of the double sesquialter ratio is in the third part of the differences, that of the double sesquitertian in the fourth, and that of the double sesquiquartan in the fifth; and so on in sequence the fractional part will always bear a name greater by a unit, when it is compared with the name of the fractional part that runs on in the species of the multiple superparticular. One should observe, in each formation both of the superpartient relations and of the multiple

superparticular ones, how an elegant reciprocity also arises. For the superpartients applied the measure in full once, and left over several unmeasured fractional parts, beginning from two: for the first is superbipartient, then come the supertripartient and the superquadripartient and so on in sequence; whereas the multiple superparticulars, reciprocally, apply the measure twice in full, but always leave over a single unmeasured part, which itself also begins from the fractional part that is yoked with the number two and advances thereafter in sequence. And in all the relations that are formed, and in those from which the formations are made, the extremes turn out to be squares.

71

The remaining mixed relation, the multiple superpartient, is generated from the superpartient. From the superbipartient, or twice-sesquitertian, specifically 9, 15, 25, when we begin from the lesser term, there is generated the double superbipartient of thirds in the terms 9, 24, 64; from the supertripartient, or thrice-sesquiquartan, 16, 28, 49, the double supertripartient of fourths in the terms 16, 44, 121; and again from the superquadripartient, or four-times-sesquiquintan, 25, 45, 81, there is generated the double superquadripartient of fifths in the terms 25, 70, 196; and continuing in succession to infinity we shall find the formation of the multiple superpartient relations advancing analogously to the superpartients and in step with them. For from the superbipartient of thirds there arose the double superbipartient of thirds, from the supertripartient of fourths the double supertripartient of fourths, and from the superquadripartient of fifths the double superquadripartient of fifths. Again, the roots of these very relations too will be seen to be arranged in good order according to a certain principle: for the roots of the double superbipartient of thirds will be observed in the fifth part of the differences, those of the double supertripartient of fourths in the seventh, and those of the double superquadripartient of fifths in the ninth, and the fractional part will always take its name by the addition

of a dyad, for instance 11 and 13 and 15, and so on always in like manner.

72

Now that we have exhibited the formation of the relations, simple and mixed, which takes its beginning from equality, a certain general theorem must be added, which will be useful to us for the ratios of harmonic theory, as follows. Each of the multiples starting from the unit, or from any prime and incomposite number whatever, will stand at the head of as many superparticular ratios of correspondingly derived name as is its own place in order from the unit or from the prime and incomposite number. For beside the first multiple in each kind one superparticular of derived name will be written in depth, beside the second in each kind two, beside the third three, beside the fourth four, and so on in sequence, so that, the diagram coming to resemble a pan-pipe, much elegance is displayed along the length, the depth and the hypotenuse.[14]

73

For from doubles there will spring both triples and sesquialters, from triples both quadruples and sesquitertians, from quadruples both quintuples and sesquiquartans, the same sequence being always preserved to whatever extent one proceeds. And the next multiple in succession will always arise along the hypotenuse, becoming a barrier to superparticulars beyond the order just stated, since it is deprived of the kind of fractional part after which the superparticular is named, as three is deprived of a half, four of a third, five of a fourth, and so on always in like manner. And in each pan-pipe the number stationed at the right angle will preserve, towards its kindred numbers on either side in breadth and in depth, a certain ratio that is not without order: in the exposition of the doubles, for example, it turns out both double and subsesquialter, in that of the triples both triple and subsesquitertian, and analogously in the rest.

73

Another theorem too must be taken up in advance, which will be most useful to us for the introduction to music, as follows. If the difference between two unequal numbers measures them according to certain other numbers that are equal to one another save for a unit, the greater according to the greater and the lesser according to the lesser, it will measure them either completely, or with excess, or with deficiency.

74

But since the complete is complete in one way only, like the perfect and the equal, in accordance with the nature of the virtues, whereas the deficient and the excessive are unlimited and indefinite, just as the vices are, on account of the nature of inequality, in the case of complete measurement the numbers measured will have one and the same ratio as those according to which the difference measured them, and the greater of the former will be to the lesser as, among the latter, the one that is greater by a unit is to the one that is less by a unit; but in the two remaining kinds of measurement they will have either a greater or a lesser ratio, and no longer the same one. Now if the measurement is deficient, so that after the measure has been applied as many times as in the case of those before them, something is left unmeasured in both of the numbers measured, and this is equal, the wholes will all come to be in a greater ratio than those parts of them that are completely taken up by the measure, when these are examined in relation to one another, and in general the numbers that lie further within and are brought down towards the less by an equal difference will always have greater and greater ratios than the larger numbers above them, just as in all arithmetic means one can see that the lesser terms are always in greater ratios and the greater terms in lesser ones; but if the measurement is excessive, so that, when the wholes have been measured out by their common difference, they overshoot the measure in their own quantity by some equal quantity, the wholes will be in a lesser ratio to one another than the numbers that mark off the equal excess beyond the measure in both.[15]

75

Let the following three pairs serve as examples of the three modes that have been mentioned: of complete measurement, 50 and 55; of deficient measurement, 48 and 53; of excessive measurement, 53 and 58; and the common difference in all of them is the pentad. Measuring, then, each of the two terms in each pair, the pentad will measure the greater ones eleven times and the lesser ones ten times. But in the first pair the wholes and the numbers according to which they were measured will have equal ratios, seeing that both alike will be in the same sesquidecimal ratio.

76

In the second pair the wholes will have a greater ratio than the numbers according to which they were measured: for the latter will be in the sesquidecimal ratio, but the wholes no longer in the same one, but in a ratio greater than the sesquidecimal; for 53 contains 48 and more than the tenth of it besides. In the third pair the wholes will have a lesser ratio than the numbers according to which they were measured: for the latter will be in the same sesquidecimal ratio, but the wholes in a ratio less than the sesquidecimal; for 58 is less than the sesquidecimal of 53, seeing that the greater contains the lesser and less than the tenth of it besides. If equal numbers are added to unequal terms, the difference of the original terms and of those with the addition will be the same, but the later ones, that is, those with the addition, will have a lesser ratio. And if an equal subtraction is made from unequal terms, the numbers remaining from them will have the same difference as the original ones, but will come to be in a greater ratio.

77

Further, this too must be taken up in advance, as it will be useful to us for the same purposes: that if any interval whatever is compounded twice, that is, if any ratio whatever is taken twice over, the middle term of course remaining common, the extremes will in every case be in a greater ratio than the terms containing the simple interval. But if

the interval that is taken twice over is in a multiple ratio, the extremes that enclose it will also be in a multiple ratio; whereas if it is in a superparticular ratio, the containing terms will be neither in a superparticular nor in a multiple ratio, but in some other, mixed relation. It is therefore possible also to state the converse, that if a compound interval has its extremes in a multiple ratio to one another, the interval that was taken twice over will in every case also be in a multiple ratio; but if the ratio of the extremes is neither multiple nor superparticular, but a mixed one, the interval that was taken twice over will not be multiple, but superparticular or of a different genus.

78

From this it will be established, in the harmonic ratios, which concordant intervals, when compounded with concordant intervals, will produce greater concords and which will not, and in what ratio the resulting composite concords stand, and in what ratio the original ones.

78

Further, this too must be taken up in advance: that a number which has multiplied another number will yield a product possessing the peculiar properties of each of its generators. And if two numbers that are in some ratio multiply two others standing in another ratio, the greater multiplying the greater and the less the less, the numbers generated from them must necessarily preserve each of the two ratios; and if the generators are roots, the rank of the ratios in question will turn out, in the products, to be that of roots as well, whereas if they are not roots, the products will preserve the same proportion of rank.

78

Likewise this too must be taken up in advance: all terms set out in a series of even number and with an equal excess, whether they be of the even nature or of the odd or indeed of both, make the sum of all the terms set out as many times the sum of the extremes alone as is the half of the multitude of the terms, from which half the multiple will take its name.

79

The topic of proportions follows next upon these matters, because proportion is a system of ratios; but postponing it for the present, we shall first go through the topic of plane and solid numbers, which belongs properly to quantity taken by itself and which was deferred for the sake of what is useful in teaching.

80

Whenever, then, any number whatever from the unit onward, either by itself or also added to those before it, is resolved into units and extended along a line, it will be called rectilinear, because it advances without breadth in length alone; for one should know that in antiquity the men of earlier times used to signify the quantities of number in a more natural way, by resolving them into units, and not, as men do now, by symbols. In the strict sense, however, those numbers are called rectilinear which do not make a plane diagram, such as 5 and 7 and the like; and they are called euthymetric because they are measured along a straight line by the unit. And since the dot is the principle and element of length, the line being, as the geometers say, the flowing of it, the monad too will by likeness have the role of dot and point, inasmuch as it is the principle of quantity and, moreover, having as it were flowed from itself and been extended in one dimension according to its own magnitude, will advance into length.

81

Thus it will also have certain attributes in common with the point: being the principle of quantity, as the point is of size; being without parts, as the point is; and having the power to produce nothing more than itself, just as the point too; for as once one is nothing more than one, so points multiplied into one another make nothing more than a point; for the line is not a composition of several points either, but either, if they touch, there will be no extension, or, if they stand apart, there will be no contact, so that the point is not, after all, a part of a line; for a point is not only that of which there is no part, but it is also not a part of anything else. The monad also has this in common with

the point: that, when it is viewed at the apex of solid pyramids whose bases have an unlimited number of angles, it is conceived, like the point, as possessing every figure. But it then has properties of its own, in which it differs from the point, being as it were of the same kind: that by composition of itself it is extended into length, and further that it is a part of this length. But if we were to cease from extension in one dimension, from drawing the units and prolonging the length, and were to pass on to breadth, arranging the units in figures upon a plane, such a number will be called plane; for it is now extended in two dimensions, and when it is drawn it is diversified in forms, beginning with the triangle; these must be gone through summarily in the following way, and an orderly generation of them must be provided.[16]

82

For when the series of successive numbers from the unit has been set out, if, leaving none out, we always add together cumulatively the successive numbers one by one, as for instance first one, then two upon this, then three upon the two and four in addition to these, and so on as far as we wish, there will be produced in succession from the unit the triangles 1, 3, 6, 10, 15, 21, 28, 36 and so on, each of which, when resolved into units, will be figured in the manner of a triangle; and so will the monad itself by itself, as being potentially triangular. Each of those after the unit will have its sides of as many units as the gnomon has, or rather, by Zeus, of as many units as the last gnomon taken up in the summation has, which is a property peculiar to triangles alone.

83

That is called a gnomon which increases each species of the polygons by addition while preserving the same species, as, for instance, the triad placed around 3, which is a triangle, produced 6, which is of the same species. The name has been transferred from geometry; for the excess by which a square exceeds a square is called a gnomon. In every case the figuration will be in the form of an equilateral triangle; so

that a triangular number would be the one generated from the numbers beginning from the unit and differing by a unit, when these are added together cumulatively. In the plane arrangement the fourth will begin to enclose the first within itself, the fifth the second, and the rest in sequence, until the seventh in its turn encloses the one that contains the first, because it is itself also fourth from the fourth; and those that follow will do the same thing analogously.

83

Again, from another starting point: if from the series of successive numbers, beginning from the unit, we add together cumulatively no longer the successive numbers but those taken alternately, that is, the odd numbers, as for instance 1, then 1, 3, then 1, 3, 5, and again 1, 3, 5, 7, and so on in sequence, squares will arise and, when resolved into units, will be arranged on the plane in the form of a square.[17]

84

The gnomons will always be placed around so as to form an angle, and the squares will no longer be increased along one side, as happened in the case of those before them. Here too, again, the third will begin to contain the first, the fourth the second, the fifth the third and also the first, the sixth the fourth and the second, and in general those of even rank the even ones and those of odd rank the odd ones. A square number, then, is the one produced from the numbers beginning from the unit and differing by a dyad, when these are added together, as 1, 4, 9, 16, 25, 36 and each one in succession, again having its side of as many units as the gnomons taken up in its composition. And since in linear figures the square, when a diagonal has been drawn, is resolved into two triangles, and plainly is also composed of these, we should find in arithmetic too a square being composed out of every two consecutive triangular numbers. The squares are also generated when each of the numbers from the unit onward multiplies itself; for the monad taken once by itself becomes square, the dyad taken twice by itself makes the square 4, and the triad taken thrice by itself makes 9, and so on in sequence.

85

If again from the series of successive numbers we add cumulatively to the unit those that leave an interval of two, pentagons will arise, 1, 5, 12, 22, 35 and so on, these too being resolved into units and figured pentagonally, the gnomons being placed around along the three sides. And again the side of each will be of as many units as the gnomons that were put together for its generation. A pentagonal number, then, will be the one produced from the numbers beginning from the unit and differing by a triad, when these are added together; a hexagonal number, the one from those beginning from the unit and differing by a tetrad; a heptagonal, the one from those differing by a pentad, and so on in sequence, the polygons having their name by an excess of a dyad over the differences of the gnomons. And if one were to set out the polygons in rows one after another, beginning from the triangle, and placing before them the continuous series of number as well, in the diagram the triangles will be seen to be even and odd alternately by twos, the squares alternately one by one, the pentagons, like the triangles, by twos, and in general those of them that are of the same rank, that is, * * * those of even rank.[18]

86

For indeed all the polygons have received their gnomons according to a certain natural good order: the triangle, gnomons alternately odd and even, one by one; the square, odd gnomons only; the pentagon again alternately, one by one; and the hexagon, odd gnomons only; and so on consistently throughout.

86

Since the monad leads off every generation of the polygons and for this reason possesses every figure, it is said in this respect to resemble a circle and a sphere, both because the circle is contained by one line and the sphere by one surface, and because the circle is able to hold and to enclose every polygonal plane figure, and the sphere every solid figure. In what follows the monad will be seen to lead off the generation of the solid figures as well, and to admit potentially the

principles of all of them; and there is the further reason that, having as it were moved from itself and around itself, it returns to itself, just as the circle too, starting from some point and moving around some point at an equal distance, returns to the same point. And if the principle of the circle shows itself in the monad, while the figurations of the polygons begin at the triad, it was with good reason that the Pythagoreans said that the dyad is indefinite, because by itself it bounds no figure whatever; for the first rectilinear figure and the element of plane figures is the triangle, because what is extended in two dimensions lies within three limits.

87

And since in linear figures the polygons are given their form by the triangle, seeing that they have their composition from it and their resolution into it, for this reason in numbers too the polygonal numbers will be given their form by the triangles according to a certain natural good order. For the potential triangle, the monad, will be the difference of the first actual polygons when they are viewed in depth, 3, 4, 5, 6, 7, 8, 9, 10 and so on; the first actual triangle, 3, which is second in order, will be the difference of the second polygons, 6, 9, 12, 15, 18, 21, 24, 27; the third of the triangles, 6, will be the difference of the third, which are 10, 16, 22, 28, 34, 40, 46, 52; and again the fourth of the fourth, and the fifth of the fifth, and so on to any extent. In the drawing of the figures of the polygons, too, two sides will in all cases remain the same, being lengthened at each step, while the sides besides these will be enclosed within by the placing-around of the gnomons and will be continually changed, one in the triangle, two in the square, three in the pentagon, and likewise to infinity, the name of the polygons here too differing by a dyad from the number of the sides that are changed.[19]

88

From this the method of the epanthema of Thymaridas was also taken. For when determinate or indeterminate quantities have shared out among them some determinate amount, and any one of them

whatever has been added to each of the rest severally, the multitude gathered from all these sums, after the subtraction of the quantity determined at the outset, is assigned in the case of three quantities as a whole to the one that was combined with the others, from which the remainder for each of the rest will then be found by subtraction; in the case of four, the half; in the case of five, the third; in the case of six, the fourth; and so on always in sequence, a difference of a dyad showing itself here too between the number of the quantities that share in the division and the name of the part. It should be observed how here too the monad has found a place for being paired with the whole; for in the theorem about the polygons that concerns the drawing of their figures we said that the side that is changed is one for the triangles, two for the squares, three for the pentagons, and so on in sequence.[20]

89

Here, however, in the epanthema, if those who share are three, then after the subtraction of the quantity that was determined we shall assign the whole of what is obtained to the one that was combined with the rest, so that here the whole corresponds to the one side that is changed in the triangles. And since there, in the case of squares, the sides that are changed will be two, here, if those who share are four, we shall assign the half, and then the third where there are five; and by always proceeding analogously we shall not go astray. That this epanthema is not superfluous, but both has reference to an arithmetical theorem and becomes for us the source of a most elegant method of discovery, we may observe in the following way. Let it be required of us, for the sake of example, to set out four numbers, of which the first together with the second shall be double the third and fourth together, and again the first together with the third triple the second and fourth together, and likewise the same first together with the fourth quadruple the two mean terms, the second and third together, and all of them together quintuple the same two mean

terms, so that the advance of the multiples from the double to the quintuple may also be in their natural order.

90

The method to be followed, then, is this. Since there is need of a half on account of the double, I take the number 2; for it is the very first that furnishes a half, and the first double. And since there is need also of a third on account of the triple ratio, I take 2 three times. The resulting 6, then, by reason of both its generators, will be the first number that admits both a half and a third. Again, since a fourth part is needed on account of the quadruple ratio, I take 6 four times, and since there is need of a quintuple, 24 five times, which makes 120; and I have this number as the common sum of the four terms, which must accordingly be posited as divisible into the four numbers that will come to light, in which the aforesaid ratios will be made manifest. The 120 is to be distributed in the following manner. Since the first two numbers will be double the remaining two, and the root of doubles is as 2 to 1, which together are 3, I take 120 twice, and I divide the 240 by 3. One part, then, comes to 80. I say, then, that the first two numbers amount to so many units, and these will be double the remaining two, which themselves of course amount to forty units.

91

Again, since the first and the third will be triple the remaining ones, the second and fourth, as 3 to 1, which together are 4, I take the same 120 three times and the result is 360, which I divide by 4, so that the part is 90. I say, then, that the first and the third together amount to so many units, being triple the remaining ones, the second and fourth, which of course amount to 30 units. Again, since the first with the fourth is quadruple the two mean terms, the second and third, as 4 to 1, which together are 5, I take 120 four times, the result is 480, I divide by 5, and I have as one part 96. So many units, then, I say, belong to the first with the fourth, and these are quadruple the two mean terms, which amount to 24 units. The numbers, then, having been found by pairing, but not yet distinguished each by itself, the knowledge of the

epanthema of Thymaridas furnishes us with a method for distinguishing them. For when the numbers of the pairs are summed together, I mean 80 and 90 and 96, the total will be 266.

92

I subtract, then, the 120 that was divided at the outset among the four terms, and there is left me 146, of which, since those that shared are four, the half will be the portion that the 80 has as its own in the first pair. The half is 73, and the remainder from the 80, namely 7, will belong to the second term. Since the second pair contains the number 90, I again subtract 73 from the 90, and there is left 17, which I say belongs to the third term. And since the third pair too is of 96 units, I again subtract the 73, and the remaining 23 I assign to the fourth term. And thus my first term comes to be of 73 units, as it were a gnomon for the discovery of the pairs, so that the four are found distinguished each individually, being in order 73, 7, 17, 23, which together are 120 and contain the stated ratios, from the double to the quintuple. These, then, are the very first and root numbers that admit the stated ratios in whole units. But if we should also be willing to divide the unit, and the odd numbers that take their form from it, into two equal parts, the halves of the numbers before us will also be seen to contain the same ratios, namely 36½ and 3½ and 8½ and 11½, whose sum too is 60, which will of course be half the former sum, 120.

93

And if we make multiples of the original numbers, according to any species of multiple whatever, or superparticulars, or superpartients, the resulting numbers will in every case contain the same ratios. But in order that, when four other numbers are set out in the same order, those of corresponding rank to the former, under the aforesaid pairing of like with like, may be generically superparticulars instead of multiples, and specifically sesquialters instead of doubles, sesquitertians instead of triples, and sesquiquartans instead of quadruples, I take by the same method, since there is need of the sesquialter ratio instead of the double, the first number able to

furnish a half, that is, 2, which was also the first double in the case of the former numbers, and I take it five times, because 5 is the sum of the numbers that contain the sesquialter ratio, 3 and 2. And since instead of the triple there is need of the sesquitertian ratio, and the root of sesquitertians is 4 to 3, together 7, I take this ten times; the result is 70.[21]

94

Again, since there is need of the sesquiquartan instead of the quadruple, and the root of sesquiquartans is 5 to 4, which together are 9, I take 70 nine times; the result is 630. This, then, will be the number that holds together the numbers containing the stated ratios. And since there is need of the sesquialter ratio, because the first two numbers will have to be sesquialter of the latter two, and the antecedent among the roots that contain the sesquialter ratio is 3, I take 630 three times and the result is 1890, which I divide by 5, which is the sum of the root terms of the sesquialter, and I get as the fifth part the number 378, which I say is the first pair, that of the first and second of the numbers that will come to light, which in the series will be sesquialter of the latter two. Again, because there is need of the sesquitertian ratio, since the first and the third number both together must be sesquitertian of the second and fourth, and the antecedent among the roots in the sesquitertian is 4, I take 630 four times, the result is 2520, which I divide by the sum of the roots that contain the sesquitertian ratio, that is, by 7, and I get as the seventh part the number 360, which becomes for me the number of the second pair, that of the first and third of the numbers that will come to light, which both together will be sesquitertian of the second and fourth.

95

Likewise, because there is need of the sesquiquartan ratio, in order that the first and fourth both together may be sesquiquartan of the two mean terms, and the antecedent among the roots in the sesquiquartan is 5, I take 630 five times, the result is 3150, I divide by the sum of the roots that contain the sesquiquartan ratio, that is, by 9,

and I get as the ninth part 350, which I say is the third pair, that of the first and fourth number, which both together will prove sesquiquartan of the second and third together. And in order that I may also separate the three pairs into the four numbers sought, I shall use the same method of the epanthema of Thymaridas. For I again sum the numbers of the pairs, 378 and 360 and 350, so that the multitude I have gathered is 1088, and again I subtract the original sum, 630. And since the terms sought are four, the half of the remaining number 458, namely 229, I assign to the first of the terms sought, which is the one that will be combined with the remaining three. And from 378, which was the number of the first pair, if I take away 229, there is left me 149.

96

This, then, I say is the second number in the series. Again, since the second pair is the number 360, I subtract the same 229 and there is left me 131, which I say is the third term in the series. Likewise, since 350 belongs to the third pair, I take away 229, I am left with 121, and I have the fourth. Of the four terms together, then, which are in order these, 229, 149, 131, 121, the first and second both together will be sesquialter of the third and fourth, the first and third together sesquitertian of the second and fourth, and the first and fourth again both together sesquiquartan of the second and third, which was to be shown. So much, then, let us have said by way of digression from outside our subject, and not without purpose, as an indication of the elegance of the arithmetical epanthemata.

96

We must return to the study of the polygonal numbers and observe how, throughout their whole diagram, it results that the successive numbers from the unit, if they have first been set out in a first row, are gnomons of the triangular row next to them, while those taken at intervals of one are gnomons of the row at an interval of one, those at intervals of two of the row at an interval of two, those at intervals of three of the row at an interval of three, and so on in sequence.

97

And all the gnomons of the heptagon will have the same endings as the first two, 1 and 6, while those of the others fall under other and yet other observations, just as in the series of the hexagon all the perfect numbers will be found, and it will be a peculiar attribute of the hexagons that they are all triangles as well, though it will no longer be an attribute of all the triangles that they are hexagons as well, but only of those at intervals of one, that is, of half of them, 1, 6, 15, 28, 45, so that here too the half may be fittingly paired with the two. For since the hexagon has come to be double the triangle in angle and in side, for this reason the triangular row will furnish from itself half its members as hexagons, and the perfect numbers in the series of the hexagons are at the same time triangles as well. And in the series of the pentagon, where there are two even numbers between the two pairs of odd numbers, one of the even numbers is necessarily even-times odd and the other odd-times even. And one who applies himself would find many other elegant consequences attending the diagram of the polygonal numbers, for instance that in depth the first numbers after the units are the numbers in succession, while the second have the triad as their common difference and are in order the superparticulars beginning from the sesquialter, and the third are superpartients, having the hexad as their common difference but being named in a certain different order in relation to one another; for they are all supertripartient, but the parts are fifths in the case of the first, eighths in the case of the next, then elevenths, then fourteenths, and so on in sequence, the parts being named always according to the half of the consequent and in conjunction with the superpartient relation.

98

More plainly, each square in the diagram is found to be the sum of the triangle above it and of the one of the same species before that, and every pentagon the sum of the triangle in line with it in depth and of twice the one before that, and every hexagon of the triangle in line with it in depth and of three times the one before that, and a

heptagon likewise of the triangle in line with it and of four times the one before that, and the same will always result, the quantity being increased by the addition of a unit.

99

Again, the second square, 9, is the sum of the triangle above it, 6, and of the one before that, 3, as has been said. The pentagon beneath it, 12, is the sum of the square above it, 9, and of the square before that, 4, less 1, one triangle lying diagonally to it. The hexagon beneath this, 15, is the sum of the pentagon above it, 12, and of the one before that, 5, less twice the same triangle, the first, 1. The heptagon beneath it, 18, is from the hexagon above it, 15, and the one before that, 6, less three times the same triangle, 1. For the first actual polygons, those ranged after the units that are polygons potentially, were less by nothing, but each was in a way from the one above it and the one before that. And again, from another starting point, the square 16, ranged in the fourth column across the breadth, is the sum of the triangle above it, 10, and of the one before that, 6, likewise less by nothing.[22]

100

The pentagon beneath it, 22, is the sum of the square above it, 16, and of the one before that, 9, less the first actual triangle, 3, which lies diagonally to it. The hexagon beneath this, 28, is composed of the pentagon above it, 22, and of the one before that, 12, less twice the same triangle, 3. The heptagon beneath this, 34, is the sum of the hexagon above it, 28, and of the one before that, 15, less three times the same triangle, 3. And so on in order the same thing will come about in like manner, the gnomonic triangles also advancing in step with the polygons taken next in order across the breadth. For the next column of polygons across the breadth, which the triangle 15 begins, will be extended in the same way as those already mentioned, according to the triangle 10; and the one after it, whose beginning is 21, according to 15. And the progression of the polygons and of the triangles that give them their form will always be extended in like

manner, so that this is a universal theorem concerning them: every polygon is the sum of the polygon above it, whose name is less by a unit, and of the triangle that stands one step lower.

101

Let the properties that belong to plane numbers, then, be taken as shown by us to this extent, as in a cursory survey.

102

Since it is now time to speak of the heteromecic numbers as well, because they too belong to the distinctive character of the plane numbers, it is right to marvel at the zeal and the precision of Pythagoras and his circle in the mathematical sciences. For those most wise men observed that all the formative principles found in number, most varied as they are and infinite in multitude, grow one and all from the monad as from some common root and pass over from potentiality into actuality: even numbers and odd, and under each of these their specific kinds, the perfect and their opposites; and further the ten relations too, fashioned from the monad; and polygonal and plane numbers from the triangle on to infinity; and further the solid numbers as well, as will be shown in what follows, in every species of solid, I mean spherical and cubic and pyramidal; and side numbers and diagonal numbers; and in short all the attributes of numbers whatsoever, as shown forth besides in the monad and made plain in accordance with it and from it; but that one formative principle alone, the heteromecic, in the whole of arithmetical theory shares with the monad in nothing, neither in receiving nor in imparting, but has been brought to light, as it were on purpose, by nature herself as something set against the monad and different in kind from it.

103

Since all the things that are are constituted in accordance with the opposition of these principles, as will be shown in what follows, the essence of harmony necessarily has its place, if indeed, according to the Pythagoreans, 'harmony is a certain fitting-together and unification of things that sound apart and are by nature at war'; and

besides, in order that here too the universal rule may be preserved, that 'there is nothing among the things that are which has no opposite'. At once, then, from the very name of otherness one can discern the opposition; for the same and the other are opposites. Now sameness and unity show themselves in connection with the nature of the monad, just as we said that it has been called monad (μονάς) because its principle possesses abiding (μονή) and rest: whether it be examined by itself, or whether it draw near to and be blended with anything else whatever, be it number or bulk or magnitude, it bestows rest and sameness upon that thing.

104

For once a hundred is 100, and once the triangle is a triangle, and once man is man, and likewise in all cases. And further, inasmuch as the monad was seen to be in a special way the form-giver of the odd numbers, and the odd numbers were seen to be gnomons of squares, and we observed that sameness and equality belong to the squares, sameness might reasonably be said to come to the other things from the monad and on account of the monad. And if sameness is in virtue of the monad, otherness will come to the things that are in virtue of the opposite power; for this in its turn will be seen to give form in a special way to the heteromecic numbers and to need the monad not at all for their fashioning, but straightway to engender in their sides otherness and deviation from the sameness that is due to the monad. For it makes the sides of every heteromecic number equal save for a unit, because it is itself equal to the monad save for a unit, and it will prove the first cause of inequality and that which displays the greater and the less. Ordinary usage, too, speaks of "the other" in the case of two things; whence the numbers that generate the heteromecic number are both two in number and differ from one another by a unit. From the same, which is also equal and like, and from the other, which is also unequal and unlike, as from two elements differing in every respect, the followers of Pythagoras held that there come to be, first of all, the properties found in numbers, on account of the

opposition of the dyad to the monad, and then, by participation in these and assimilation to them, all the things in the cosmos as well.[23]

105

For all other things manifestly imitate number, while number has from itself, as its principles, the monad and the dyad. Just as, then, narrowing the name down from all the figures that have four sides and four angles, we called "square" (τετράγωνον) the one that has all its sides and angles equal, so too, out of all the species of otherness in respect of the sides, we shall call "heteromecic" the number that displays the deviation of otherness at its very nearest, that is, the one that has the other in its lengths to the extent of a unit, being so named in contradistinction to the number that is the same in length. Euclid, once again failing to perceive this, confounded in this case too the differentiation and variety of the theory, supposing that the heteromecic is simply the number that arises from two different numbers multiplied together, and not distinguishing the oblong from it.

106

Yet if one were to grant him this, it will follow that opposites, which by nature cannot exist together, are found at the same time and in the same thing: for his account declares the same number to be square and yet also heteromecic, 36 for instance and 16 and many others, which would be equivalent to the odd number's being the same as the even. But if the odd and the even are arrayed in good order alternately one by one, by nature herself and not by our convention, and could never be confounded, so too the squares and the heteromecic numbers, being most natural, will themselves observe good order, inasmuch as they have their fashioning and arrangement from those, the monad leading and ruling the odd numbers, and the dyad the even. For from 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, and so on as far as one likes, added together, there arise the squares 1, 4, 9, 16, 25, 36, 49, 64, 81, 100; and from 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 the heteromecic numbers 2, 6, 12, 20, 30, 42, 56, 72, 90, 110. And the former, being equal

times equal, will have as their sides the successive numbers from the unit, while the latter, being unequal times unequal in the nearest degree, that is, save for a unit, will have the successive numbers from the unit taken two by two, selected in the conjunct manner, so that their sides too may differ from one another by a unit.

107

In the generation of the squares, then, the monad carries off the title of cause of their constitution. For in the placing-around of the gnomons it is the monad that is there beforehand, and without it the adding-together of the odd numbers by themselves would not generate squares; and in the heaping-up of the successive numbers after the manner of the so-called double course the monad offers itself as both starting-gate and finishing-post in each addition. For from it the advance takes its beginning in the generation of each square, as from a starting-gate, up to the side of the square that is to be produced, as to a turning-post, and back to it again comes the return, as to a finishing-post, all the numbers, and the monad itself, being taken twice, except the turning-post, which will also be the side of the square that corresponds to it. For in this way it will come about that each of the numbers, receiving the advance from the monad as far as itself, and making from itself the recoil of the backward run toward the monad, is the side of a square: 2 the side of the square 4, for 1 and 2 and, on turning back, 1 again make the square 4.

108

And 3 is the side of 9: for 1 and 2 and 3 and, on turning back, 2 and 1 make the square 9. And the fourth, 4, is the side of 16: for 1 and 2, 3, 4 and, on turning back, 3, 2, 1 make the square 16. And let anyone test this as far as he pleases: he would find that all the numbers within the last number, which is the side of the square, are taken twice in the addition, both in the advance from the monad and in the return to it, while the side number alone is not taken twice, and has the role at once of beginning and of end and, besides these, of mean: of beginning, because from it comes the return to the monad; of end,

because to it comes the advance from the monad; and of mean, because it bounds both the advance and the return, being as it were a turning-post. And perhaps it is for this reason that the whole sum-total of the numbers added together in advance and return is its power (δύναμις), since, stationed alone as upon an acropolis, it is escorted by the remaining numbers in their advance as by an armed force (δύναμις). In the constitution of the heteromecic numbers, on the other hand, if one has to place the heap of the even numbers around something in the manner of gnomons, the dyad alone will be seen to receive and sustain the placing-around, and without it no heteromecic numbers will grow.

109

And if the successive numbers are heaped together after the same double course, the monad, as being the principle of all things according to Philolaus (for 'One', he says, 'is the principle of all things'), will offer itself in like manner to the heteromecic numbers too as a starting-gate for their generation, but it will no longer also be the finishing-post of the backward run and return upon turning; rather the dyad will take up that part in its stead, for it is to the dyad itself that the return will belong. Now the advance from the monad up to the two side numbers, which in the case of the heteromecic numbers will have the role of turning-posts, resembles a coming-to-be that proceeds from the common principle of all things to the turning-posts themselves as to its prime, while the return from these, being a kind of dissolution and decline, resembles a perishing. Hence it is reasonable that the monad, as having the role of form, will lend itself to the constitution even of the heteromecic numbers themselves, but no longer to their dissolution and, as it were, perishing; the close, rather, will be in the dyad, which has the role of matter, just as we see in natural things too that all the things in the realm of becoming have their coming to be, and each one's being a "this" and being one, by reason of form, but their perishing and not being, and being indefinite instead, by reason of matter.

110

For the "this", when deprived of form and shape, would be matter, indefinite and without quantity and without quality, on account of the indefiniteness and inequality of the dyad. For this reason the dyad was seen to be in a special way the form-giver of the heteromecic numbers, imparting to them in their sides its own power, that is, inequality, for the unequal is twofold, excess and deficiency; while the monad is the form-giver of the squares, which is why they are equal times equal; for the one and the monad are the principle of equal things, if indeed the equal is one to one, and equal things are equal in virtue of a single ratio. It is clear, then, that all the things in the cosmos have been constituted and come to be out of form and matter analogously, just as all the properties found in number come from monad and dyad. For primarily each of the two principles is the form-giver of one of the two lengths of number, I mean the even and the odd, and secondarily the one of squares and the other of heteromecic numbers; and their powers do not overlap, but, being most contrary, each disposes the things that partake of them according to its own proper character. For just as the hot naturally heats the things that come near it, and the cold cools, and the moist moistens, so too the principles of the things that are, being unmixed with the other powers, shape all the things that partake of them according to their own proper powers.

111

Now it is the nature of the one and the monad to define, to limit, to give shape, to equalize, to preserve and in general to unify, and of the dyad to divide, to cut in two, to destroy and in general to render indefinite; and that is why, in the generation of the heteromecic numbers just described, the monad will no longer lend itself to the constitution of the dyad itself, but the dyad by itself, as being itself also a principle, is at once the root of the heteromecic numbers. And the reason why it could not come from a principle is that, as Plato says, it would then no longer be a principle. Analogously, among the

cosmic principles too the demiurgic god is found not to be generative of matter, but to have taken it over as itself eternal, moulding it and making it into a cosmos by forms and ratios in accordance with number. To the constitution of the remaining heteromecic numbers, however, the monad will contribute itself in the advance only, as we said, and no longer in the return as well; in this way, for example, from 1 and 2 and 3 arises 6, the heteromecic number next in sequence after the dyad, having as its sides the dyad and the triad, although both of these have the role of turning-posts.

112

For in the squares, on account of the sameness and equality of the sides, there turned out to be a single turning-post, which was the side-number of each square; but here, in the heteromecic numbers, because the sides must be different and unequal, two turning-posts were needed, and we are prevented by the 6 from adding on any number by way of return, since the monad is laid down as not admitting of return and resolution; and the dyad is a turning-post no less than the triad, but both alike, with equal authority, are side-numbers of the heteromecic number 6, producing it from twice 3 or from thrice 2. In all the heteromecic numbers the turning-posts must be taken once only, in the advance alone, as we did in the case of the squares as well. Again, from 1, 2, 3, 4 and, in the turning back, 2 alone, arises 12, the third heteromecic number, whose sides are the two turning-posts, 3 and 4; for 12 is produced by 4 times 3.[24]

113

Moreover, from 1, 2, 3, 4, 5 and, in the turning back, 3, 2, arises 20, the next in regular order, which likewise has the two turning-posts as its sides and is generated from 4 times 5 or 5 times 4; and this will come about without end by the same procedure. The heteromecic numbers too, then, like the squares, will have a varied generation: by composition, by blending, and by the double course already described. By blending: just as the squares arose from once 1, twice 2, thrice 3, four times 4, and so on as far as one likes, so the heteromecic numbers

will arise from once 2, twice 3, thrice 4, four times 5, and so on in order, two numbers that differ from each other by a unit being blended in pairs. By composition: just as the squares were first one odd number, then two, then three, then four, and so on always in the same way * * * no longer by pairing but by addition to those that were there from the beginning. Their generation by the so-called double course has been spoken of a little earlier. The aforesaid formation of each species is said to be 'by blending' because the number produced can no longer give back its gnomons unmixed, owing to their fusion; rather, when they are separated out, they show through one another. For instance, 6, which comes from twice 3, is not resolved into 2 and 3, but the fusion has produced something more than the quantity of the gnomons.[25]

114

For each of the gnomons is present in the number generated as many times as its partner has units, and for this reason they are said to show through one another, just as happens with liquids that are blended, juices and things poured and things melted and the like; for separation into the original ingredients is not possible, because the qualities have been fused together and show through one another. The other formation is said to be 'by juxtaposition and composition' because the numbers produced can be resolved into those from which they were compounded; 6, for instance, composed of 2 and 4, can be divided into the same, so that every multitude too that is composed by heaping or by herding together can be separated into its single units. Alone of all numbers the dyad, as we learned earlier, makes the product by blending equal to the product by composition, whereas the numbers after it make the product by blending greater than that by composition, and the monad before it, conversely, makes it less. That is why the followers of Pythagoras, from what happens in its case, called it 'equal' and 'just'; and from a fact of this kind its seminal and principle-like character is recognized. For as the monad * * * and contains in the manner of a seed, undiscriminated, the formative

principles that are in number, so too the dyad alone will contain, confused and undifferentiated, the peculiar property of blending and that of juxtaposition, which will not belong even to the monad, but will be peculiar to the dyad.[26]

115

In natural things too we should find that all seeds hold the formative principles of what is to be brought to completion from them undiscriminated and confused, inasmuch as they are potentially those things which will come to be from them.

115

Again, then, from another starting-point: since the squares are powers of certain numbers increased by their own length, while the heteromecic numbers are increased not by their own length but by another, it was not unreasonable that they were called 'heteromecic'; and in contradistinction to this it would not have been unfitting to call the squares 'idiomecic'. The ancients called the squares 'same' and 'like', on account of the likeness and equality in their sides and angles, and, by contrast, called the heteromecic numbers 'unlike' and 'other'.

116

In the setting-out of each species, the squares will be alternately odd and even, one after the other, because numbers of that kind are what increase them; but the heteromecic numbers are all even, because an odd number lengthens an even one or an even an odd, and every odd number multiplied by an even one generates an even number.

116

And since we have reached this point in the discussion, it should be known that this example will be useful to us for the nuptial number in Plato's Republic, where he says that from two good parents there will in every case be good offspring, and from two of the opposite kind the opposite, while from mixed parents there will in every case be bad offspring and never good. For from the coming together and successive addition of the odd numbers by themselves, with the monad leading, there arose squares, which are of the nature of the

good as coming from parents of that kind; and the cause of this is equality and, prior to it, the one. From that of the even numbers, with the dyad leading, there arose heteromecic numbers, which are of the opposite nature, because their begetters are so too; and the cause of this, again, is inequality and, prior to it, the indefinite dyad. And if there should be a mixture and, as one might say, a marriage of even and odd, the offspring generated are also of the nature of the other, whether the begetters differ by a unit or by some greater number; for the numbers produced are either heteromecic or oblong.[27]

117

And again, when squares are mixed with one another the numbers that result are squares, and when heteromecic numbers are, the results are like them, but from mixed parents the results are never squares but in every case of a different kind; and this, says the most divine Plato, is what the rulers of his Republic, men and women alike, will overlook, either because they have not been reared in the mathematical sciences or, even if they have been so reared, through heedlessness, and so will mingle the marriages indiscriminately; and the offspring of these, turning out inferior, will become the beginning of faction and dissension for the whole polity.

117

But in order that we may also learn the harmonious and most closely cohering conjunction of the two species, squares and heteromecic numbers, although they are of the most opposite nature, we must set out each of them in a row, parallel to one another, starting from its proper principle, the squares from the monad and the heteromecic numbers from the dyad, thus:

117

1 4 9 16 25 36 49 64 81 100

117

2 6 12 20 30 42 56 72 90 110

118

And one must observe how the first of the 'other' numbers, in relation to the first of the 'same', contains the root ratio of the first of the multiples; the second in relation to the second, the first of the superparticulars in a form derived from its root; the third in relation to the third, the second of the superparticulars in a form derived from its root; and the fourth in relation to the fourth, the third of the superparticulars in a form derived from its root; and anyone who examines this as far as he wishes will find it advancing in regular order. The difference of all of them from all, when they are examined pair by pair, will be the successive numbers from the unit. When the rows are examined each by itself, in the case of the like numbers the differences will be the odd numbers from the triad on, and in the case of the unlike numbers the even numbers from the tetrad on. And again, each difference of the unlike numbers taken two by two will have a superparticular ratio to the corresponding difference of the like numbers, and the ratios will in every case be named from odd numbers: sesquitertian, sesquiquintan, sesquiseptiman, sesquinonan, and so on in sequence. Again, the number made up from the first like number, twice the unlike number beneath it, and the second like number is a like number; and that which arises from the third like number, twice the unlike number beneath it, and the fourth like number is a like number; and if we go on always in this way, so that the end of the earlier generation begins the later one, we shall generate nothing but like numbers.

119

If, conversely, we begin from the unlike numbers, placing them as extremes and the like numbers as means in each conjunction, all the numbers generated will be unlike and of the nature of the other. But if we insert as means not the like numbers of middle rank but those that come next in order at each generation, keeping the same unlike numbers as extremes, the results will be the like numbers that were passed over, 16 and 36 and 64 and those analogous to them. And these are all even, because the like numbers inserted as means, even if they

are odd, make even numbers when they are taken twice together with the unlike extremes, which are even; for every odd number taken twice becomes even. But the former were all odd, because one of the two like extremes was in every case odd, and through being taken once they preserved the oddness. The conjunction of the gnomons in these same cases displays certain well-ordered ratios: from once the first like number, twice the first unlike number and once the second like number, the subdouble ratio will arise; from the second like number, twice the unlike number beneath it and the next like number, the subsesquialter; in the third conjunction the sesquitertian, in the fourth the sesquiquartan, and so on in sequence.[28]

120

And in the generation of the like numbers that were passed over, the conjunction of the gnomons no longer displays the three terms of each pairing in the same ratio, but in different ones; not, however, in unrelated ones, but again with a certain natural good order and kinship, of the double ratio to the sesquialter, of the sesquialter to the sesquitertian, and of the sesquitertian to the sesquiquartan. For in the terms 2, 4, 6 there are the double and the sesquialter ratio; in 6, 9, 12 the sesquialter and the sesquitertian; in 12, 16, 20 the sesquitertian and the sesquiquartan; and so on analogously, the second ratio being named, in comparison with its partner, from a number greater by a unit. Again, each like number together with the unlike number of the same rank makes a triangular number; and the triangular numbers so produced, beginning with 3, will be every other one, namely these: 3, 10, 21, 36, 55, 78, 105, and so on analogously, leaving out from the regular formation of the triangular numbers 6 and 15 and 28 and 45 and 66 and 91 and those analogous to these. But if we use not only the composition of parallel terms but also that by interweaving, entwining the first unlike number with the second like number, the second unlike with the third like, the third with the fourth, the fourth with the fifth, and so on continually, then together with the former ones all the triangular numbers from the triad on will arise in order,

namely these: 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, and those that follow without limit.[29]

121

Again, the halves of the unlike numbers themselves, taken by themselves, will make the triangular numbers in regular order from the unit. And each difference between unlike and like numbers, taken pair by pair, will have a ratio that is not disorderly to the numbers of which it is the difference: for it will be the half of the one and the third of the other, then the third of the one and the fourth of the other, then the fourth of the one and the fifth of the other, and so on continually. The beginning of this good order will be furnished by the second pairing, that of 4 to 6; for the first pairing, that of 1 to 2, will not have this feature, because the one, and the monad, which has the role of form and sameness, is without parts. The dyad will be the first to admit of division and separation, being of the nature of the other and having taken on the role of matter; and since, being paired with the monad, it was prevented on account of the latter from beginning the said good order of the parts, it is itself found as the difference in the second pairing, being the half of 4 and the third of 6.

122

But when the dyad is compared with 4 as well, it none the less keeps this difference in relation to it. And since the three terms 2, 4, 6 do not differ in quantity, that of their differences, they do differ in quality, that of their ratios; for 4 is the double of 2, but 6 the sesquialter of 4. The same 6, however, when compared in like manner with the next number, 9, will not differ in quality; for it will preserve the same sesquialter ratio, presenting itself as the consequent, just as in relation to 4 it was the antecedent of the same ratio; but in quantity, that of the difference, it will differ, since in relation to 4 the difference is a dyad, but in relation to 9 a triad. Again, 9, when compared with 6 and also with 12, will differ in the quality of its ratios, since it is the sesquialter of the one and the subsesquitertian of the other, but in quantity, that of the differences, it will not differ; for its difference in

relation to each of them is a triad. And in general, wherever three terms taken in the way described differ in the quantity of their differences, they will be without difference in quality, that of their ratios; but if they should differ in quality, they will be without difference in quantity.

123

The like and the unlike numbers may also be recognized from one another: for the first unlike number is made up of twice the first like number, and the second like number is made up of twice the first unlike number, while the second unlike number is made up of one and a half times the second like number. Again, the third unlike number is made up of one and a third times the third like number, just as the fourth like number is made up of one and a third times the third unlike number. The fourth unlike number is made up of one and a fourth times the fourth like number, even as the fifth like number will be made up of one and a fourth times the fourth unlike number, and the fifth unlike number will be made up of one and a fifth times its partner, and the sixth of one and a sixth, and the same thing will always come about in due sequence, the part being named according to the quantity of the place of each of the unlike numbers when it is compared with the like number of the same rank, of which the part will also primarily be a part, and secondarily a part of the unlike number too, when this is compared with the next like number. And one might find many other elegant things by gazing intently at the diagram by oneself and continually examining the harmonious relation of the opposites, the two powers of sameness and otherness that are displayed in the setting-out of the squares and the heteromecic numbers.

124

A sufficient encomium of the decad will be the generation of the squares according to the double course already described, when, in the first rank of numbers, which the decad itself bounds, the advance is made from the unit up to the decad, and again from it, as from a

number that marks off units from tens, the return is made back towards the unit; for from the decad, as by composition, there will arise the square number 100, which is itself also a joint marking off tens from hundreds and is called by the Pythagoreans 'unit of the third course', just as the decad is called 'unit of the second course' and the thousand 'unit of the fourth course'. The side of the square 100 will be the decad itself, and its power will be the sum total of the heaping up upon it of the numbers within it, taken twice; for it was in this way that the manner of the successive addition of the numbers, both that by advance, from the beginning as from a starting-gate, and that by return, from the end as from a turning-post, was said to have been likened to a double course.

125

But if we should use the decad no longer as a turning-post but as a starting-gate and beginning of the advance as far as the hundred, from which the return will again be made to the decad, from the successive addition there will arise the number 1000, the unit of the fourth course, which is itself too a joint marking off hundreds from myriads. But the hundred will no longer be also the square side of the number one thousand; for one thousand is not a square at all, but a cube, from the decad as its side. But in order that it may be made plane in the manner of an oblong, its side will be the hundred together with the decad as well, so that it is clear that the hundred will need the decad in order to become a side. Again, if we should use the hundred as a beginning and in place of a starting-gate, and should go forward adding in succession the hundreds after it up to the thousand, and from this, as from a turning-post, should likewise return to the hundred as to a finishing-post, the result will be the number ten thousand, the unit of the fifth course, which has as its side, as a square, the hundred, but, as an oblong, the thousand together with the same decad.[30]

126

Thus the decad, in order to become itself a side in the manner of the double course, will need none of the other joints of number, I mean the hundred and the thousand; but these, in order that such a thing may befall them, will in every case need the decad, which is why we have assigned this encomium to it.

126

It remains to mention also all the other properties that can be thought out by those who, out of love of contemplation, bend their efforts to the discovery of the attributes of numbers: for instance, that every square either has a third straightaway, or, if it does not, at all events has a fourth, or, if not this either, then, when a unit is subtracted, from a square that has a third you will produce a number that has a fourth, and from one that has a fourth a number that has a third, and, if it has neither, a number that has both; and if it should have both, there are times when the subtraction of the unit deprives it of both.

127

And every number, when it has multiplied either one of the numbers of the same kind that differ from it by a dyad on either side, and has taken on a unit, makes a square. Odd numbers make even squares, and even numbers odd ones. And every number, when it has multiplied a multiple of itself, will make a number that is the same multiple of the square produced from itself, and likewise if it takes a superparticular or a superpartient or a mixed one. Similarly also every triangular number, when it is taken eight times and has taken on a unit, makes a square; and when two squares are multiplied by one another the number produced is a square; and of numbers in continued proportion from the unit, if the one next to the unit is a square, the rest too will be squares; and when any three numbers are in proportion, if the first is a square, the third too will be a square; and when a square measures a square, side too will measure side; and every number produced by multiplying together the sides of two consecutive squares will be a mean proportional between them; and many other such things we

shall both find out by ourselves, if we are eager, and be able to learn by inquiry as they have been worked out by others.

127

For the present, however, we must pass on to the ratio of side and diameter, which has received very ample examination in geometry, because it is in accordance with it that the figures seem in some way to be regulated and given their form.

128

Just as, then, we did in the case of the figures themselves, transferring their ratios by likeness to the numbers as well — for those things too are made rational by numbers — so also in discussing side and diameter we must follow the nature of number and preserve the likeness as far as is possible. For it is not the case that, just as among sizes, when the side has been made rational the diameter is irrational, or conversely when the diameter has been made rational the side is irrational, so it is among quantities also; rather, the side will be rational in relation to the diameter, in order that number may be rational in every way and may have this as its special prerogative, inasmuch as it is most of the nature of a principle and has become the cause of rationality for all other things. For it is common to numbers and magnitudes, inasmuch as they are incorporeal, to be unmoved, but it is peculiar to number not to admit incommensurability either, whereas magnitudes do admit it. We must, then, once again trace methodically from the monad the generation of the ratio of side and diameter, since we said that the monad takes the lead of all the ratios that are in numbers.

129

For we must name two units, the one side and the other diameter, and employ certain universal additions that are always the same, adding to the side a diameter and to the diameter two sides, since in linear figures what the side produces in power twice, the diameter produces once. The diameter, then, becomes greater than the side by a unit. But at the outset, without the addition, the square in power on the unit

diameter is less by a unit than the double of the square in power on the unit side; for the units, being in equality, make the one less by a unit than the double of the other. But when the addition has been made as described, the square on the diameter will be greater by a unit than the double of that on the side; for they are 9 and 4. Again, if we add to the side a diameter and to the diameter two sides, they will be 7 and 5, and the square on the diameter becomes less by a unit than the double of that on the side; for it is 49 to 25. Again, if the same addition should be made, the square on the diameter will be greater by a unit than the double of that on the side; for it is 289 to 144.

130

And so in like manner, as the addition is made on the same principle, the square on the one will be at one time greater by a unit than the double of the square on the other, and at another time less by a unit, and in this way sides and diameters become rational in relation to one another. But since, alternately, the diameters are in power at one time greater by a unit than the double of the sides, and at another time less by a unit than the double, all the diameters together will be, in conception, double in power of all the sides together; for an equalization comes about when the greater is mingled with the less, because equality is the coming to rest of what exceeds in relation to what is exceeded; wherefore here too that which is greater by a unit than the double, when it is added to that which is less by a unit than the double, will equalize the whole, so that the diameter is always double in power of the side, just as is proved in the case of linear figures as well. And let so much be said by us about the attributes of plane numbers.

131

A solid number is one that has taken on a third dimension in addition to the two in plane numbers, evidently when a fourth term has been added; for what is extended in three ways lies in four terms, so that, there being one that takes and one that is taken and a third according to which it is taken, it is itself a fourth. Of solid numbers, then, some

are equiangular and equal in their faces and equal in their dimensions, these too being taken by likeness to those among linear figures; and these are called cubes and tetrahedral pyramids, whose base may be exchanged in every direction; others are parallelepipedal and equiangular but unequal in their dimensions, the species of which are bricks and beams; others are unequal in their faces and unequal in their angles and unequal in their dimensions, and are called stakes or, as some say, altars or wedges, each name having been given by likeness; and others are mixed, having all their angles equal save one and again all their faces equal save one, namely pyramids, those which begin from the one that uses a square base and go on to infinity, in which there will no longer be an exchange in respect of the base, as came about in the case of the one that uses a triangular base.

132

Among plane figures, what is properly called a square among quadrilaterals corresponds to the cube, the parallelogram to the brick or to the beam, which some call a pillar, and the trapezium to the wedge. An example of the cube, which is extended in every direction equally an equal number of times, is 8 and 27 and 64 and 125 and 216, which arise from twice two twice and from three times three three times and four times four four times and five times five five times and six times six six times. While all of these are called cubes, let those which at every advance end in the same thing be called, more particularly still, spherical as well, since they have been increased by one more dimension from circular numbers, which themselves too have the same ending, as 125, which is from the side five, and 216, which is from the side six. And even if these are increased still further, each of the two will none the less end in its own side.

133

The monad, just as it embraced all the forms found among plane numbers apart from the heteromecic principle, so also embraces those found among solids; for it will be pyramidal, being observed at the apex of every species of pyramid and holding in each case, potentially,

the role of a solid point; for the angles of every solid number will be point-like units, potentially greater than those in plane numbers, because they are solid; for the point is simple, being the limit of the magnitude extended in one dimension, but potentially double in planes, on account of the convergence of the two lines upon one point, and in solids potentially indefinite, beginning from triple, because the first convergence, that of three sides, produces a solid angle, the pyramidal one. Moreover the monad will be spherical, just as it was also circular, being extended three times according to its own magnitude. Of the numbers unequal in every dimension let 60 serve as a common example; for it arises from three times four taken five times, and conversely from five times four taken three times, and from four times five taken three times, and from four times three taken five times.

134

Of parallelepipedal numbers, examples of bricks, which are equal times equal a lesser number of times, are 18, which is from three times three taken twice, and 48, from four times four taken three times; and of beams, which some call pillars, and which are equal times equal a greater number of times, 36, which is from three times three taken four times, and 45, from three times three taken five times; for it is possible both in these and in the bricks to make the diminutions and the increases not only adjacent, that is, by single units, but also at an interval, so that the likeness of configuration may be the more apparent. The account of pyramids would become easier and readily worked through if we were to draw out the exposition of the polygonal numbers, beginning from the triangles, in parallel rows as we did a little earlier, and then to fit those of the same kind upon one another in a heap, in good order, as far as any number whatever, so that in each piling the apex is in every case a unit, and for every one the base proves to be potentially of like figure with it. Through the triangles 3, 6, 10, 15, 21 and so on, then, there will arise the pyramids having a triangular base, namely these: 4, 10, 20, 35, 56; through the

squares 4, 9, 16, 25, 36, those employing a square base, 5, 14, 30, 55, 91; and through the pentagons 5, 12, 22, 35, 51, those employing a pentagonal base, 6, 18, 40, 75, 126.

135

We shall do the same in the case of the subsequent polygons as well; for just as we had as gnomons of the polygons the successive numbers from the unit, so too we have as gnomons of pyramids the successive polygons of each kind. The number of the planes, too, will correspond to the sides of the gnomons; and just as among the polygons those of odd rank were even and odd two by two, and those of even rank one by one, so also among the pyramids, in those of odd rank there is one odd pyramid alternating with three even ones, and it ends, moreover, in a pentad, except for the potential pyramid; for they stand in fifth places from one another; while those of even rank go two by two, those which here too have the like ending necessarily coinciding with the odd pyramids among those of odd rank. Each pyramid is composed of the pyramid of different species above it and of that one, among those lying in the plane, which stands one step lower, as was the case with the polygons as well: for example, 5 is composed of 4 and 1, 6 of 5 and 1, 7 of 6 and 1, and again 14 of 10 and 4, 18 of 14 and 4, 22 of 18 and 4; and going on in sequence, fitting them together along the depth and the breadth of each diagram of the polygons, we shall find analogous results, namely that each pyramid is composed of the one above it and of the one beneath that one; for first there is no difference, then a difference of once into the plane, then of twice, then of thrice, and so on.

136

And the other properties too we shall find, in the same manner and analogously, in the case of these as well. In breadth they will differ from one another by their own bases, while in depth, after the row in equality, which is set out in a straight line, the difference will be the tetrad, which is the element of pyramids in actuality, then the decad, the second pyramid, then the number twenty, the third pyramid, and

so on in sequence. If a pyramid does not come to its apex in a unit but in the gnomon next to it, it is called truncated; if not even in that one, but in the next, twice-truncated; and likewise it will be named thrice-truncated and four-times-truncated and so on always in sequence, according to the number of the gnomons taken away.

137

We shall find many peculiar properties of cubes as well, just as of squares; for when each of the numbers from the unit onward multiplies itself and then the product arising from itself, the cubes arise in good order. And if the squares from the tetrad onward, in order, should multiply the successive numbers from the dyad onward, in order, each its counterpart, or be multiplied by each, the cubes will likewise arise in good order. Further, since the odd numbers are productive of likeness and are of the nature of the same, as was shown, if they are added together in selections, always with the addition of one more, cubes will be produced: for example, first 1, the potential cube, incomposite; then two odd numbers, 3 and 5, give 8, the second cube; then three odd numbers, 7, 9, 11, give 27, the third cube; then four, 13, 15, 17, 19, give 64, the fourth cube; and similarly with those that follow. Again, in the exposition of numbers in proportion the third terms are squares, the fourth cubes, and the seventh at once cubes and squares. Every cube, when multiplied by its own side, makes a square, which will be as many times the cube as the square on the cubic side is of the side itself; and this latter square will itself be the square side of the number produced from the cube and its side.

138

Again, as from two squares multiplying one another a square arose, so from two cubes there arises a cube, and from a cube taking itself a number that is at once cube and square. And among numbers in proportion, if the one after the unit is a cube, the rest too will be cubes; and when four numbers are in proportion, if the first is a cube, the fourth too will be a cube; or again, when a cube measures a cube,

the side too will measure the side. And practically all the attributes of squares will be observed, analogously, in cubes as well. Leaving, then, the discovery of such properties to those who will pursue these refinements for themselves, we shall pass on to the topic of proportions.

138

Proportion, then, is the likeness and sameness of several ratios. What ratio in the sense that concerns proportion is, since the word is used in many senses, we made clear in what went before: it is the relation to one another of two homogeneous terms. 'Homogeneous' is added because it is proper to compare things that fall under the same genus, for example a mina with a talent, whose common genus is weight, and a line with a surface or a solid; for magnitude is common to them.

139

There are also some things compared in respect of power and in respect of bulk, and certain other genera. But how things that are not homogeneous stand to one another it is not possible to know, for example a cubit to a cotyle, or the white to a choenix. Quantity too is one genus, and number belongs to quantity, so that comparison will arise in the case of the ratios in number as well: they will have a certain ratio and a relation of a certain kind. If the terms are in equality, the ratio is that of equal to equal; for equality is without difference; but in inequality it is according to difference. And interval will not be the same thing as ratio: the ratio is twofold, because the unequal too is two and not one, and the interval will be the same but the ratio different; for of two to one and of one to two the interval is the same, but the ratio is double and half, so that ratio is something other than interval; for indeed in the case of several terms, while the ratio is often the same, the interval is different, as in 4, 6, 9. That the ratio of inequality is in ten genera, five of them antecedent ratios on the side of the greater and an equal number consequent ratios on the side of the less, and that all have their generation from equality, we learned earlier in the topic of relations.

140

There is also a ratio called by this very name, that 'of number to number', because it falls under none of the ten genera, as will be shown in the harmonics: the ratio of the leimma, in the terms 256 to 243. The ratios in numbers being, then, of some such kinds, proportion will be the taking together of several ratios in likeness, in three terms at the least; for a ratio is said to be conjoined when there is a common mean term having a ratio to each of the extremes; for it is the common term that joins the ratios. A ratio is said to be disjoined from a ratio when they have no common term. This occurs in four terms, which is why 'the proportional' is thought to differ from 'proportion'; for the proportional occurs in disjoined terms as well, whereas proportion in the strict sense is assigned to those that have a common term.

141

Since proportion, then, arises in three terms, the first term must have to the second the ratio which the second has to the third, or conversely, which is also why it has been so named (ἀναλογία); for the terms are set out according to the same ratio (ἀνὰ τὸν αὐτὸν λόγον). Their differences too will be in the same ratio; and if there is ratio in equality also, it is clear that there is proportion as well. Of this the most elementary is that in units, so that there may be a proportional monad also, then that in dyads, and third that in triads, and so on in sequence; from these, in accordance with the three rules stated earlier, the proportions in inequality are generated in good order.

141

It must be understood beforehand that the ancients gave the name proportion in the strict sense to the geometric, though by now, more loosely, all the rest as well are so called, which generically are means. That the name was with good reason restricted to the geometric will be stated in the account of it. In ancient times there were only three means, in the days of Pythagoras and the mathematicians of his time: the arithmetic, the geometric, and the one third in order, which was

once called subcontrary but was afterwards renamed harmonic by the circle of Archytas and Hippasus, because it manifestly embraced the ratios that belong to what is attuned and melodious.

142

It was formerly called subcontrary because it underwent something subcontrary to the arithmetic, as will be shown. When the name had been changed, the later mathematicians of the circle of Eudoxus, having discovered three further means, called the fourth subcontrary in a special sense, because it too undergoes something subcontrary to the harmonic, as will be shown; the remaining two they named simply from their order, fifth and sixth. The ancients and those who came after them thought it possible to construct just so many means, that is, six; but the more recent writers discovered four others besides, contriving their generation out of the terms and the intervals.

142

The first, then, is the arithmetic mean, when the mean term has an equal interval to the extremes on either side, and exceeds and is exceeded by an equal number, but has different ratios to the extremes, the greater ratio being that to the lesser term and the lesser that to the greater, and these ratios being continuous though of different genera.

143

As an example of it: when the series of successive numbers from the unit is set out and any three terms whatever are taken, whether consecutive, or skipping one, or skipping two or three or four or as many as one wishes, the mean term in each selection exceeds the lesser and is exceeded by the greater by an equal number, for example 1, 2, 3 and 1, 3, 5 and 2, 4, 6. It is generated from equality in this way: a first term equal to the first, a second equal to the first and second, a third equal to the first and second and third; or again, a first from the first and second, a second from the first and two seconds, a third from the first, two seconds and the third. But from the equality in units, by the former procedure, there is generated the mean that has its terms

with none omitted; from that in dyads, the one skipping one; from that in triads, the one skipping two; and in tetrads, the one skipping three, and so on analogously.

144

And if the first ratio is double, the second is in every case sesquialter, and that of the extremes triple. If it is triple, the second is superbipartient of thirds, and that of the extremes quintuple. And if it is quadruple, the second is supertripartient of fourths, and that of the extremes septuple, and so on analogously. It is a property of this mean that the mean term is the subdouble of the two extremes together. And again, as each term is to itself, so also is the excess to the excess; and this is the same as the terms being at an equal excess. The combinations taken from the unit by three terms will produce the polygonal numbers that are second in actuality, all exceeding one another by a triad; for from 1, 2, 3 there arises the second triangular number in actuality, 6; from 1, 3, 5 the second square in actuality, 9; from 3, 4, 5 the second pentagonal in actuality, 12; and so on in sequence. But if the terms are selected from the unit skipping one each time, the triangular number will no longer begin the polygonals, but the lead will pass over to the square; for the first will be 9, from the combination 1, 3, 5, and those that arise next will observe a certain principle that is not without order.[31]

145

If the selection is made with an omission of two, so as to be 1, 4, 7, the pentagonal number 12 will begin. If with an omission of three, there will be from 1, 5, 9 the hexagonal number 15, and so on without limit, in conformity with the generation of the polygonal numbers themselves. For because the triangular numbers were produced from the numbers with none skipped, in the first constitution it is the triangular number 6 that will begin the polygonals; and because the squares were produced from the numbers skipping one, in the second constitution the square 9 takes the lead; and further the pentagonals from those skipping two; and this will hold throughout in sequence.

Now since the first combination from the unit with none skipped is productive of a hexad, this first combination, 1, 2, 3, will give their form to those that follow it, when no term is taken in common nor yet any passed over, but after 1, 2, 3 there is taken 4, 5, 6, then 7, 8, 9, and so on in sequence; for all these will turn out to be hexads, the decad always taking over the place of a unit, that is, being reduced to a unit; for it is in this way, as we were saying, that it is called by the Pythagoreans a unit of the second course, and the hundred a unit of the third course, and the thousand a unit of the fourth course.

146

For 4, 5, 6 makes the number 15; and when the decad is reduced to a unit, the five, taking this in addition, becomes a hexad. Again, 7, 8, 9 added together makes the number 24; reducing its 20 to two units I add them to the 4, and again I have a hexad. Again, adding together 10, 11, 12 I make 33, of which the 30 is a triad; adding this to the three I likewise have a hexad, and this will be so in the same way throughout. And the first hexad has no transposition of a decad into a unit, inasmuch as it is the form-giver and element of those after it; but the second will have a transposition of one unit, the third of two, the fourth of three, the fifth of four, and so on in sequence. And as many as are the decads transposed, so many enneads also will be subtracted from the whole sum, so that what is left may likewise be a hexad; for since 15 has a transposition of one decad, if I take away one ennead, a hexad will be left. And since 24 has two decads that are converted, if I take away two enneads, again a hexad will be left, and this will turn out so throughout.

147

One might find still more elegant properties attending the arithmetic mean, which we deliberately pass over for the present, having regard to the due measure proper to an introduction. This is the mean that Plato described as 'exceeded by an equal amount in number, and exceeding by an equal amount'.

147

The second mean, the geometric, is called proportion in the proper sense, because its terms contain the same ratio, being separated according to the same ratio; for whatever ratio the terms have to one another, whether from the lesser to the greater through the common term or the reverse, this the difference also has to the difference; and the reason is that the terms will not be separated by an equal difference, as in the former mean. It is also possible for the proportional to occur in four terms, the ratios being disjoined. And, so that we may fit Plato's saying to the proportion here, we must quote it: 'For whenever, of three numbers, whether bulks or powers, the middle one has something in common, so that what the first is to it, that it is itself to the last, and again conversely, what the last is to the middle, the middle is to the first, then, the middle becoming first and last, and the last and the first in turn both becoming middles, it will follow that these are thus of necessity the same.'[32]

148

Before Plato, too, Pythagoreans had made the same determinations about it. Thus Timaeus the Locrian, in his On the Nature of the Cosmos and the Soul (from which, they say, Plato furnished himself and composed the Timaeus that for this reason bears his name; among those who say so is Timon, the author of the Silloi, who speaks as follows: 'For much silver he got in exchange a little book, and setting out from there he tried his hand at Timaeus-writing'), says something like this: 'For of any three terms whatever, whenever the intervals too stand in the same ratio to one another, then indeed we see that the middle, by the measure of justice, is to the first just what the third is to it, and conversely and alternately.' Geometric proportion belongs to continuous quantity, that is, to size, being separated according to equal and similar ratios; whereas the arithmetic belongs to discrete quantity, being separated no longer by equal ratios but by equal numbers in its excesses. And in the latter the ratios are different but the intervals the same; in the geometric, conversely, the ratios are the same but the differences different.[33]

149

This mean too is generated from equality, by the same three rules that were used in the case of the relations; for all the sets of three terms there are in proportion according to this mean, standing thus: as the greater is to the mean, so is the mean to the lesser, and so is the excess of the greater over the mean to that of the mean over the lesser. It is a property of it to make the product of the extremes equal to the square of the mean term, if the terms are three or in general odd in number; but if they are four or in general even, it will make the product of the extremes equal to the product of the means. And in this mean the terms lengthen one another by blending, but in the arithmetic by composition, because such is the nature of discrete quantity and multitude, with which, again, arithmetic is properly concerned, as we said at the beginning of the introduction. Now among multiples, in proportional series of every kind, we shall find a great many examples of it, but among superparticulars and superpartients ever rarer ones, in step with the advance of the name of the part.

150

The reason is plain: every number is capable of being multiplied, but not every number can admit all parts; rather, those that skip one admit halves, those that skip two thirds, those that skip three fourths, those that skip four fifths, and so on, the numbers that have parts of greater denomination being ever more sparse. And if the ratios will be ever fewer because of the rarity of the numbers that will admit the part by virtue of which they will become superparticular or superpartient, much more will the proportions become rarer because of the addition of the third term; for a number that contains, besides the mean term, let us say, a half of it as well, does not in every case itself have a half, nor does one that contains some number together with its third part itself also have a third, and similarly with the parts that follow. But in order that a proportion may arise, it is necessary that the root terms which contain the ratios multiply one another, and these same terms will also show themselves in the differences of

the proportion. And so that we may take a common example of root proportions in all the species of the superparticular, beginning from the sesquialter, and in addition in the first of the multiples, that is, the double, we must here too set out in rows the numbers that are same and those that are other, each kind from its own beginning, and fit them together by interweaving, so that each combination is of three terms, and in continuity, the last term of the preceding combination always beginning the one after it; for by this undisjoined selection each set of three terms from the unit will exhibit what is sought.

151

The third mean, the one called harmonic, exists when, of three unequal terms, as the greater term is to the least, so is the excess of the greater terms to the excess of the lesser, that is, the excess of the greater over the mean to that of the mean over the lesser. This mean differs from those before it in that the mean term neither exceeds and is exceeded by the extremes by an equal number, nor stands in the same ratio to them. Its roots are 2, 3, 6 or 3, 4, 6; for by the multiplication of these, or by their superparticular increase, if indeed they admit of it, many others will spring up. Some call this mean stationary, because it appears only in the aforesaid root terms, as being stationary and prototypical; for in the case of the arithmetic and the geometric it is possible to arrange an unlimited number of combinations.

152

At any rate, in both the root means the extremes are in double and in triple ratio to one another, and so are the differences of the greater terms from the means to those of the means from the lesser. The mean is called harmonic because the ratios found in harmony can be seen in it in seminal form: thus in 3, 4, 6 the concord called the fourth, which is the least of the concordant intervals, being observed in sesquitertian ratio, is in the lesser terms, that is, in 4 to 3; the fifth, which comes next after the fourth, being in sesquialter ratio, is in the greater terms, that is, in 6 to 4; and the octave, being composed of

both the aforesaid and observed in double ratio, is in the extremes, that is, in 6 to 3. Moreover, the difference of 6 from 4 to that of 4 from 3 is likewise in double ratio, in accordance with the concord of the octave.

153

Furthermore, the power of the extremes multiplied into one another, 18, in relation to that of the mean multiplied into itself, 16, being in sesquioctave ratio, contains the interval of a tone; for in the prototypical terms 3, 4, 6 it was not possible for the ratio of this interval to appear, because none of them is able to furnish an eighth part, by virtue of which some other number would be its sesquioctave. Again, the power of the greatest is triple, and the triple ratio contains the concord of the octave and fifth together; and the power of the greatest by itself to the power of the least will have quadruple ratio, which contains the concord of the double octave. And again, starting afresh: the power of the least into the mean is 12, of the same into the greatest 18, and of the mean into the greatest 24; while the power of each on its own is, of 3 by itself 9, of 4 16, and of 6 36. And in sesquitertian ratio, the one that contains the fourth, are 24 to 18 and 12 to 9; in sesquialter, that of the fifth, are 18 to 12 and 24 to 16 and 36 to 24; in triple ratio, so that the octave and fifth may be constituted, 36 to 12; in quadruple, so that the double octave may appear, 36 to 9; and in sesquioctave, for the manifestation of the interval of a tone, 18 to 16, as was said before.

154

The other root mean, 2, 3, 6, has the triple ratio immediately, both in the extremes to one another and again in the differences to one another, in which ratio is the mixed concord of the octave and fifth, something that did not belong to the former mean, 3, 4, 6. And if we should also multiply the terms by themselves and into one another, and the differences by themselves and into the terms and further into one another, more ratios of concords will spring up for us, as anyone

who pursues such refinements for himself can perceive. Plato's saying too may be appropriately fitted to this mean.

155

For the harmonic mean is the one 'exceeding and exceeded by the same part of the extremes themselves', which is an attribute of no other mean: for in 2, 3, 6 the mean term exceeds and is exceeded by the same part of the extremes, that is, by a half; it exceeds the lesser and is exceeded by the greater; and in 3, 4, 6 again the mean term, by the same part of the extremes, a third, exceeds 3 and is exceeded by 6; for it does so by a unit and by a dyad. This mean was held by Pythagoras and his school to be subcontrary to the arithmetic mean, because the latter had its mean term exceeded and exceeding by a part of its own, no longer of the extremes, and by the same part; for it exceeds and is exceeded by an equal number or unit, whereas in the harmonic it is not by an equal amount. But since some wish to take this mean as subcontrary to both, the arithmetic and the geometric, whereas we said that it undergoes something subcontrary to the arithmetic alone, the following consideration too will come to our aid: for what will stand in their way is that the geometric mean is seen to undergo something mixed and to hold the position of a mean in relation to both the arithmetic and the harmonic, as to what is in each case an extreme.

156

For it will blend in itself the peculiar properties of each of the two. For it was peculiar to the harmonic that the mean term exceeds and is exceeded by a part of the extremes themselves which is the same in quality, even if not in quantity, but never by a part of the mean term; and to the arithmetic, conversely, that it is no longer by a part of the extremes but of the mean term, and by a part the same in quantity. In the geometric, however, the part by which the mean term exceeds and is exceeded belongs neither to the extremes alone nor to the mean term alone, but both to the mean and to the extremes; for it will exceed one of the extremes by a part of itself, and will be exceeded by

the other by a part of that extreme; and the part will be the same in quality, even if not in quantity, as in the harmonic. Often, too, it will exceed and be exceeded by several parts, again the same in quality, so that it will also have something in common with the harmonic, namely that the part is the same in quality only and no longer in quantity; and in this respect the harmonic will not be subcontrary to it. Again, in the arithmetic the lesser ratios appeared among the greater terms and the greater among the lesser; in the harmonic, in the subcontrary manner, the greater among the greater and the lesser among the lesser; while in the geometric, as being in a way a mean between them, they are neither lesser nor greater, but equal.

157

For these reasons, then, the harmonic might reasonably be called subcontrary to the arithmetic alone, and not to the geometric as well. The harmonic has as its property that the product of the mean term and the sum of the two extremes is double the product of the extremes alone. It is generated, once more, from equality, first from units, then from dyads, then from triads and so on, by the following rules: the first term from the first and the second; the second from the first and two seconds; the third from the first, twice the second and three times the third, so that there may arise the mean which has its extremes and its differences in triple ratio. But if it is to be in double ratio: the first term from the first and twice the second; the second from twice the first and twice the second; the third from once the first, twice the second and three times the third. For from equality in units there will arise, according to the rules stated, the two root means 2, 3, 6 and 3, 4, 6; from equality in dyads their doubles, from equality in triads their triples, and so on. And upon all the formations that arise the peculiar properties of the harmonic will follow.

157

Just as on the canon, while the fastenings remain fixed, the movable bridge by being shifted produces a variety of concords, in the same way it is possible, when two terms are given, whether even or odd,

and remain the same, to produce one mean after another, now the arithmetic, now the geometric, now that which is subcontrary to the arithmetic, that is, the harmonic.

158

For observe: with even terms, 40 and 10, the term 25 when made the mean produces an arithmetic mean, upon which all its peculiar properties will follow; 20 produces a geometric mean together with its peculiar properties; and 16 a harmonic mean with the attributes that belong to it. With odd terms, 45 and 5, the same 25 inserted as mean will likewise produce the arithmetic mean.

159

The reason is that by as much as the greater term has gained in relation to the mean, by so much has the lesser been diminished, so that the mean again exceeds and is exceeded by an equal excess; for this was the property of the arithmetic mean. And 15 inserted as mean will produce a geometric mean, and 9 the harmonic. Numbers smaller than the extremes here set out, in the odd species and in the even, which are capable of containing the three means, one would not find; these would be the root numbers and the least.

159

These three means alone were deemed worthy of account by the ancients, having the differences from one another and the properties in themselves that have been stated, and they were also fitted by them to the constitution and harmony of the cosmos, as we shall show elsewhere. The three that come after these were themselves also deemed worthy of acceptance from the time of Archytas and Hippasus; and the first of them, which is the fourth when it is counted together with the original three, is, as we said, called 'subcontrary' as its proper name, because it undergoes something subcontrary to the harmonic, on account of the ratios of the concords that have been observed in the latter. The fourth mean, then, is of this kind: of three terms, as the greater is to the lesser, so will the difference of the lesser terms be to that of the greater.

160

It is said to be subcontrary to the harmonic because in that mean the difference of the greater terms was the antecedent, whereas in this one it is that of the lesser. Both preserve the same extremes, both in their root forms and in the multiples of these. Examples of it will be 2, 5, 6 and 3, 5, 6; and its property is that it makes the product of the greater terms a multiple of the product of the lesser. The mean term will exceed and be exceeded neither by the same part of the extremes themselves, as in the harmonic, nor by a part of itself, as in the arithmetic, nor at once by a part of itself and of one of the extremes, as in the geometric; rather there will be a certain property peculiar to this mean in respect of the excess. This mean too is generated from equality, primarily that in units, then that in dyads and triads and so on in sequence: the first term from the first and the second; the second from the first, two seconds and two thirds; the third from once the first, twice the second and three times the third; and there will arise the mean that has its extremes in triple ratio.

161

In order that the one having them in double ratio may arise, one must make the first term from the first and two seconds; the second from the first, two seconds and two thirds; the third from once the first, twice the second and three times the third. The prototypes just stated, then, will arise from the equality of units, their doubles from that of dyads, their triples from that of triads, and so on in sequence. The fifth also undergoes something subcontrary, in its case to the geometric, but it is called simply 'fifth' because the name had already been claimed by the mean before it. It exists, then, when of three terms, as the mean is to the least, so is the difference of these same terms to that of the greater, for example 2, 4, 5. It stands in subcontrary opposition to the geometric because in that mean the difference of the greater terms was a multiple of that of the lesser, whereas in this one, conversely, that of the lesser is a multiple of that

of the greater, in the same ratio, however, as that of the terms mentioned.

162

It has as its property that it makes the square on the mean term double the product of the least and the mean, and further the product of the greater terms double the product of the extremes. The multiples of this mean too will have the same consequences. It is generated from three equal terms: the first term from the first and the second; the mean from two firsts and two seconds; the greater from the first, two seconds and two thirds. The sixth exists when, as the greatest of the three terms is to the mean, so too is the excess of the lesser terms to that of the greater. For the same reasons as the one before it, it is opposed to the geometric; for example 1, 4, 6. Here too the difference of the lesser terms is the antecedent, whereas the geometric has as antecedent that of the greater terms; and the ratios both of the terms to one another and of the two differences to one another are sesquialter. Of the same kind too will be the ratios that render the property of this mean. For the square on the greatest term, 6, is sesquialter of the product of the greater terms, and this same product is sesquialter of the square on the mean. This mean too may be generated from three terms in equality, as follows: the first equal to the first; the second to two firsts and two seconds; the third from once the first, twice the second and three times the third.[34]

163

The prototype, then, is from units, and its multiples from dyads and triads and the equalities that follow.

163

We have now spoken also of the three means that come next after the first ones, which were employed by those from Plato down to Eratosthenes, their discovery having been begun, as we said, by Archytas and Hippasus the mathematicians. As for the four that were ingeniously added by the more recent men who came after these, Myonides and Euphranor the Pythagoreans and their circle, it is

neither right to pass them over, for that would show a want of love for what is fine, nor yet to prolong the account of them, since they possess nothing so august or so varied as those before them. We must therefore speak of them cursorily, keeping in view at the same time the due proportion of the book. We have named them simply thus: seventh, eighth, ninth and tenth.

164

The seventh exists when, as the greatest term is to the least, so is their difference to that of the lesser terms, for example 6, 8, 9. Its generation is from the fourth, 3, 5, 6: for adding together the extremes of that mean I set the sum as the greatest term of this one; from the least and the mean I make the mean of this one; and the greatest of that one I make the least of this. It is a consequence of this mean that the oblong contained by the greater terms has to that contained by the lesser the same ratio as the greatest term has to the least, and as the difference of the extremes has to that of the lesser terms. The eighth is observed when, as the greatest is to the least, so too is the excess of these same terms to the excess of the greatest over the mean, conversely to the one before it, for example 6, 7, 9.

165

The generation of this one too is from the fifth, 2, 4, 5: for adding together its greater terms I make the greatest of this one; the extremes I set as the mean of this one; and adding together in turn the lesser terms of that mean I have the lesser term of this one. It is a consequence of it that, as the greatest is to the least, so too is the product of the greater terms to the product of the lesser. The ninth exists when, as the mean term is to the least, so is the difference of the extremes to that of the lesser terms, for example 4, 6, 7.

166

Its peculiar property is that the product of the greater terms stands to the product of the extremes in the same ratio as the mean term to the least, and as the difference of the extremes to the difference of the lesser terms. We shall generate this one too from the sixth, 1, 4, 6: for

by adding together its extremes I make the greatest term of this one, as mean I set the greatest term of that one, and as least its mean term. Thus the three means from the seventh onward will have their generations, in order, from the three before them, the fourth, the fifth and the sixth. The tenth, coming after them all, is when, as the mean term is to the lesser, so also is the difference of the extremes to the difference of the greatest from the mean, for example 3, 5, 8. The peculiar property of this one is that it is observed in superpartient ratio, and in its root form at that, but not in multiple or superparticular ratio.

167

And it has as a consequence that it makes the oblong contained by the lesser terms equal to that contained by the said differences; the reason is that they are the same numbers. This one too is generated from the harmonic mean 2, 3, 6, which stands before the three means just mentioned, so that consecutive means may have their generation from consecutive means, even if not from ones in regular order. Adding together, then, the extremes of that one I make the greatest term of this one; out of the lesser terms I set the mean of this one; and the mean term by itself I keep as the least of this one. Now that ten means in all have come to light for us, this too will be no ordinary encomium of the decad, bearing as it does on the fact that no perfect ratio escapes it, but that, being as it were a kind of receptacle (dechas), it receives into itself the ratios of all the things that are, and for this reason was named by the ancients All and Whole and Heaven, as we shall try to show in the account devoted to it, when, immediately after this introduction, we set out also the special properties (ἐπανθήματα) of each of the other numbers from the unit up to it.

167

For the present we must speak of the most perfect proportion, which exists in four terms and has been specially named musical, because the musical ratios of the concords in harmony are most clearly contained in it.

168

They say that it was a discovery of the Babylonians and that it first came to the Greeks through Pythagoras. Many of the Pythagoreans, at any rate, are found to have made use of it, such as Aristaeus of Croton, Timaeus of Locri, Philolaus and Archytas of Tarentum, and a good many others, and after them Plato in the Timaeus, where he speaks as follows: 'After this he went on to fill up both the double and the triple intervals, cutting off yet further portions from that source and placing them between these, so that in each interval there were two means, the one exceeding and exceeded by the same part of the extremes themselves, the other exceeding by an equal amount in number and exceeded by an equal amount; and, sesquialter and sesquitertian intervals having arisen, he filled them up throughout with the leimma of the sesquioctave'; and so on with what follows these words, all of which will be clear after the exposition of this proportion. The proportion called musical, then, is in four terms, two extremes and two means, in such a way that the ratios of the mean terms to the extremes, being different, are interwoven in accordance with the concords that stand at their intervals in the harmonies.[35]

169

For since the concords in music come about in harmony when two or even more notes not of the same pitch are blended under a single stroke and fall upon the hearing as one in form, the least and first concordant interval perceptible to the hearing is the fourth (διὰ τεσσάρων); for the strings that contain it are at that distance from one another. This is in sesquitertian ratio; and after it, when one string has been added, the whole interval is for the same reason called the fifth (διὰ πέντε), and this concord too is in a ratio, the sesquialter. The difference of the latter from the former is the interval contained by the added fifth string, which is that of a tone and is in sesquioctave ratio, so that the fifth will exceed the fourth by a tone, and the sesquialter ratio the sesquitertian by a sesquioctave. And these have been apprehended as incomposite intervals and as simple among the

concords; the greater intervals compounded out of them render the concord by now more full-bodied, and first among them the so-called octave (διὰ πασῶν), which is compounded of both of those and has itself received this name because it embraces all the strings that produce the simple concords; and it is in double ratio, for the double is the combination of every sesquitertian and sesquialter ratio.

170

From the double itself, again, and each of the two original ones come the octave together with the fourth and the octave together with the fifth. What is called the octave together with the fourth the Pythagoreans did not think to be a concord, since it escapes multiple and superparticular ratio, and superpartient as well, and falls out into a mixed relation; for it is as 8 to 3, because 6 is the double of 3 and 8 the sesquitertian of 6. For the present, however, let it too be reckoned a concord, in accordance with the more recent writers, for the sake of clarity in what follows. After it, again, the octave together with the fifth is a concord, being in triple ratio, because the triple ratio is composed of the double and the sesquialter: for 6 is the double of 3, and 9 the sesquialter of 6, and 9 stands to 3 in triple ratio.

171

The double compounded with itself makes the concordant interval of the double octave, which is in quadruple ratio; for the double ratio taken twice is quadruple. The 5 concords greater than this come about when the original simple intervals are again interwoven with the double octave; these we deliberately pass over for now, since it is more opportune to give a technical account of them in the Introduction to Music itself. As for the tightenings and slackenings of the strings that take place according to the said ratios, they relate that Pythagoras was the first to measure them out: for, having gone into a smithy and caught the sound of a concordant resonance from the downward blow of the hammers, he weighed the weights against one another, and finding them * * * and in the said ratios, with great natural genius he thought the matter through and fitted the same ratios to a variety of

materials, now to the lengths of strings either equal in thickness but measured against one another in respect of their shortening, or conversely equal in length but proportionally thickened, now to strings that did not differ in the aforesaid respects but were measured out differently in tension alone, and often to strings that underwent the test in respect of two or even three of the said differences.[36]

172

And from there it was very easy for him to apply what is analogous to pan-pipes and auloi and wind instruments in general as well: for there too, in conformity with the stringed instruments, the lengths and the bores, when measured out according to the said ratios, produced the concords, the width and length of the pipes corresponding to thickness, length and slackness of string, and narrowness and shortness to thinness, tension and shortness. The causes on account of which this came about we shall make clear in the proper place, in the Introduction to Music itself. For the present we must consider the said ratios in numbers, as in a cursory survey. In order, then, to set off a sesquitertian ratio, one needs a number that has a third; for its sesquitertian will in every case have a half, so that the sesquialter of this latter too may be double in relation to the original number, as is the case with 6, 8, 12. Or again, in order to make a sesquialter ratio, I need a number that has a half, so that its sesquialter, when extended, necessarily having a third, may furnish a subsesquitertian ratio in relation to some other term, which will again be a multiple of the original number, as is the case with 6, 9, 12.

173

If, then, we keep the extreme terms fixed, since they are in fact the same in both means, I mean 6 and 12, and set the mean terms together between them, there will be the aforesaid musical proportion in four terms, 6, 8, 9, 12. We set the number 6 as its first term, since it was the first to admit of a third and a half together, in order that we might be able to set off from it the said ratios: 8 as sesquitertian, in relation to its sesquialter 12, which will be double the original 6; or again 9 as

sesquialter, in relation to its sesquitertian 12, which again will be double the original number. And this is the aforementioned interweaving of the mean terms with the extremes. That it was necessary to set 6 first for the setting-off of the ratios we may learn from the following: the unit was not possible, since it is posited as partless and has neither a half nor a third; nor yet the dyad, because it has a half but has no third; nor indeed the triad, which has a third but has no half; nor the tetrad, for the same reasons as the dyad (for it too is deprived of a third); and the pentad is excluded for the same reasons as the unit, having neither a half nor a third. So 6 will be the first and least number to serve us for the settings-off of the ratios, being the product of the first two numbers that admit, the one of a half and the other of a third, I mean 2 and 3.

174

In relation to 6, then, 8, being its sesquitertian, will contain the concord of the fourth. And since 12 is sesquialter in relation to 8, containing the fifth, the same 12, being double in relation to the original 6, will contain the concord of the octave, which is compounded of the fourth and the fifth. And again 9, being the sesquialter of 6, will contain the fifth, and 12 in relation to 9 the fourth, being its sesquitertian, and in relation to the original 6 once more the octave. The excess of the fifth over the fourth, the interval of a tone, will be contained by the mean terms, 9 in relation to 8, which are in sesquioctave ratio, because the sesquialter ratio exceeds the sesquitertian by a sesquioctave, as has been said. Moreover the difference of the greater terms, 12 and 9, in relation to that of 8 and 6 will contain the sesquialter ratio, which belongs to the concord of the fifth; and, by interweaving, the difference of 12 and 8 in relation to the difference of 9 and 6 will have the sesquitertian ratio of the concord of the fourth; and the double ratio will be had by the difference of 8 and 6 in relation to the difference of 8 and 9, and by the difference of 12 and 8 in relation to the difference of 8 and 6, and further by the

difference of 12 and 6 in relation to the difference of 9 and 6; for in each pairing the double is the ratio of the concord of the octave.

175

The triple ratio, for the concord of the octave and fifth, will be contained by the difference of 9 and 6 in relation to the difference of 9 and 8, or also by the difference of 12 and 6 in relation to the difference of 8 and 6. And the quadruple ratio, that of the concord of the double octave, by the difference of 12 and 8 in relation to the difference of 9 and 8. One might find still more ratios of the concordant intervals by multiplying the four terms set out, each by itself and by one another, and further by their differences, and the differences themselves both by themselves and by one another, as each person can come to see by trying it for himself, out of love of what is fine. Once again, then, there is a musical mean when, in four terms, as the greatest is to the one next to it, so the lesser of the mean terms is to the least; and as the greatest is to the lesser of the mean terms, so the greater of the mean terms is to the least. It has as a consequence that it makes the oblong contained by the extremes equal to the heteromecic number produced by the mean terms. And it alone contains the three primary means as well: the arithmetic mean in the terms 12, 9, 6, for the mean term exceeds and is exceeded by an equal amount; the harmonic in the terms 12, 8, 6, for the mean term exceeds and is exceeded by the same part of the extremes themselves; and the geometric in disjunction, for as 12 is to 8, so is 9 to 6, so that the sameness of the ratios is rendered through the 4 terms.[37]

176

Let this, for the present, be the end of our Introduction according to Nicomachus the Pythagorean; on another occasion, god granting, we shall compose and provide for you this same Introduction to Arithmetic itself in more complete form, as for one who has by then acquired through the present one a disposition capable of following it; and all the other things that flower upon the numbers from the unit to the decad, according to the physical account and the ethical and,

prior even to these, the theological, we shall set in order and study together with you, so that from them the exposition of the three introductions that come next, I mean the musical, the geometrical and the spherical, may thereafter become easier and most ready for you.

Notes

[1] καταλήγουσαν is read for the printed καταλέγουσαν (the allusion is to Epinomis 991e).

[2] τεχνῶν τε καὶ ἐπιστητῶν: translated as ἐπιστημῶν. The clause τὸ δὲ ἀριθμητικὸν ἓν ἰδίων οὐχ ὑποπεσεῖται οὖν ... is corrupt and is rendered by its apparent sense.

[3] κωνική ('conical') is translated as printed; κυκλική is expected beside σφαιρική and ἀποκαταστατική.

[4] The μέν-clause on the monad has no answering clause; a sentence on the dyad has probably fallen out.

[5] μείζονα: the sense requires μείονα ('smaller'), which is translated.

[6] ἀδιάλυτα is read for the printed διαλυτά, as the contrast with the greater extreme requires.

[7] ἐπίπλαστος is read as ἐπὶ πλάτος; διπλάσιοι ἄρτιοι περισσῶν as διπλάσιοι ἀρτιοπερίσσων (12, 20, 28 being the doubles of 6, 10, 14).

[8] The correction of Euclid's definitions is damaged; Pistelli supplies 〈περισσάρτιος, ἀ〉 μφότεροι, and a clause on the even-times even seems to have dropped out.

[9] Lacuna in the text after πρὸ τοῦ εἰς, marked * * *.

[10] ἄσχιστος ('undivided') is read for the printed ἄσχετος; compare Nicomachus I.17.4.

[11] ὅρον ('term') is read for the printed λόγον.

[12] ἐν ἐπιμερεῖ (the roots of the superpartient) is read for the printed ἐν ἐπιμορίῳ, as the contrast shows.

[13] τὸ ἔλαττον ('the less') is read for the printed τὸ ἴσον.

[14] ἁπλῶν ('simple') is read for the printed ἁπλατῶν.

[15] The definition of deficient measurement does not fit the example that follows (48 and 53); translated as it stands.

[16] ἀπειρογώνων and ὁμογενής are read for the printed ἀπειρογόνων and ὁρογενής.

[17] The words εἴρηται δὲ γνώμων ὁ, omitted in the First1KGreek transcription, are restored from Pistelli's print.

[18] Lacuna after τουτέστιν, marked * * *.

[19] περίεισιν is corrupt; translated by the sense (the common difference is 6).

[20] προσνέμει τ' ἀφ' οὗ is corrupt; the sense is taken from the parallel on the next page.

[21] Pistelli's supplement 〈ὑποπολλαπλάσιοι γίνωνται〉 is translated as printed, though the species that follow (sesquialter, sesquitertian, sesquiquartan) are superparticulars.

[22] παρὰ τὸν ε′: the arithmetic (12 = 9 + 4 − 1) and the next clause require α′, which is translated.

[23] τοῖς ἄλλοις is read for the printed τοῖς ἀλόγοις.

[24] In Pistelli the sentence ends ὅ τε γ′ καὶ ὁ δ′, ιβ′ τετράκι γ′ ἀποτελεῖται; the First1KGreek transcription displaces its last words, which are here put back.

[25] Lacuna after καὶ ἀεὶ ὁμοίως, marked * * *.

[26] Lacuna after ὡς γὰρ ἡ μονάς, marked * * *.

[27] οἱ γεννώμενοι ὄγκοι καὶ τῆς καθ' ἑκατέρου φύσεως is corrupt; translated as the following clause requires.

[28] The statement that unlike extremes with like means give unlike numbers is translated as transmitted, though that construction yields the even squares named in the next sentence.

[29] γνωμόνων is read for the printed γενομένων, and ἐμπλοκήν for ἐμπλοπήν.

[30] ὁ πρῶτος ἀριθμός is read as ὁ ͵α ἀριθμός (1000).

[31] ἐκ δὲ τῆς γ′ δ′ ε′ is printed; 1, 4, 7 is expected (so p. 145). The sum is 12 either way.

[32] αἴτιον δέ τι is translated as αἴτιον δ' ὅτι. The Timaeus quotation differs from Plato's text and is translated as given.

[33] The Doric quotation from Timaeus Locrus is unsound; ποτ' ἄλλα is read as ποτ' ἄλλαλα and κἂν πάλιν as κἀνάπαλιν.

[34] Pistelli's supplement 〈τὸ〉 ἀπὸ τοῦ μέσου: the arithmetic (36 : 24 : 16) requires a genitive, and the sense is translated.

[35] The quotation of Timaeus 36a–b is abridged and garbled at its end; translated as transmitted.

[36] τὰς δὲ ε′ τούτῳ μείζονας: the numeral is suspect; translated as printed. Lacuna after εὑρόντα, marked * * *.

[37] The words περιέχει δὲ καὶ τὰς πρωτίστας τρεῖς μεσότητας μόνη, omitted in the First1KGreek transcription, are restored from Pistelli's print.